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YOLO advances to its genesis: a decadal and comprehensive review of the You Only Look Once (YOLO) series

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Abstract
This review systematically examines the progression of the You Only Look Once (YOLO) object detection algorithms from YOLOv1 to the recently unveiled YOLOv12. Employing a reverse chronological analysis, this study examines the advancements introduced by YOLO algorithms, beginning with YOLOv12 and progressing through YOLO11 (or YOLOv11), YOLOv10, YOLOv9, YOLOv8, and subsequent versions to explore each version's contributions to enhancing speed, detection accuracy, and computational efficiency in real-time object detection. Additionally, this study reviews the alternative versions derived from YOLO architectural advancements of YOLO-NAS, YOLO-X, YOLO-R, DAMO-YOLO, and Gold-YOLO. Moreover, the study highlights the transformative impact of YOLO models across five critical application areas: autonomous vehicles and traffic safety, healthcare and medical imaging, industrial manufacturing, surveillance and security, and agriculture. By detailing the incremental technological advancements in subsequent YOLO versions, this review chronicles the evolution of YOLO, and discusses the challenges and limitations in each of the earlier versions. The evolution signifies a path towards integrating YOLO with multimodal, context-aware, and Artificial General Intelligence (AGI) systems for the next YOLO decade, promising significant implications for future developments in AI-driven applications.

Keywords You Only Look Once · YOLO · YOLOv1 to YOLOv12 · YOLO configurations · CNN · Deep learning · Real-time object detection · Artificial intelligence · Computer vision · Healthcare and medical imaging · Autonomous vehicles · Traffic safety · Industrial manufacturing · Surveillance · Agriculture

1 Introduction

Object detection is a critical component of computer vision systems, which enables automated systems to identify and locate objects of interest within images or video frames (Liu et al. 2020; Badgujar et al. 2024; Ahmad and Rahimi 2022; Gheorghe et al. 2024; Arkin et al.

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2023). Real-time object detection has become integral to numerous applications requiring real- and near-real-time analysis, monitoring and interaction with dynamic environments such as agriculture, transportation, education, and health-care (Fernandez et al. 2016; Wang et al. 2018; Ren et al. 2015; Tang et al. 2024; Chen and Guan 2022; Ragab et al. 2024). For instance, real-time object detection is the foundational technology for the success of autonomous vehicles and robotic systems (Flippo et al. 2023; Malligere Shivanna and Guo 2024; Flores-Calero et al. 2024), allowing the system to quickly recognize and track different objects of interest such as vehicles, pedestrians, bicycles, and other obstacles, enhancing navigational safety and efficiency (Guerrero-Ibáñez et al. 2018; Shoman et al. 2024a). The utility of object recognition extends beyond vehicular applications, and is also pivotal in action recognition within video sequences, useful in digital surveillance, monitoring, sports analysis, cityscapes (Hnewa and Radha 2023) and human–machine interaction (Hussain and Zeadally 2018; Fernandez et al. 2016; Shoman et al. 2024b). These areas benefit from the capability to analyze and respond to situational dynamics in real time, illustrating its broad applicability, acceptance, and impact. However, the problem of object detection involves several challenges:

- Complexity of real-world environments: Real-world environments/scenes are highly variable and unpredictable. Objects can appear in various orientations, scales, distances and lighting conditions, making it difficult for a detection algorithm to generalize and maintain accuracy in real time (Kaushal et al. 2018).
- Illumination factors: Illumination plays a crucial role in object detection, as factors like lighting intensity, direction, shadows, and glare can significantly affect performance (Xiang et al. 2014; Xiao et al. 2020). Non-uniform or low light, color temperature changes, and dynamic lighting variations can obscure object features or cause false detections. Solutions include controlled lighting setups, preprocessing techniques like normalization and color correction, and training models with diverse, augmented datasets to enhance robustness (Seoni et al. 2024).
- Occlusions and clutter: Objects may be partially or fully obscured by other objects, creating cluttered scenes that result in incomplete information, which requires careful interpretation for accurate analysis (Khan and Shah 2008; Mostafa et al. 2022).
- Speed and efficiency: Many applications necessitate rapid processing of visual data to enable timely decision-making. This requires detection algorithms to achieve a balance between high accuracy and low latency, ensuring that the systems can deliver efficient and reliable results in real- or near-real-time scenarios, such as autonomous vehicles and traffic safety, healthcare and medical imaging, industrial manufacturing, security and surveillance and agricultural automation (Gupta et al. 2021).

Addressing these challenges, as discussed below, required innovative techniques, which conventionally relied on hand-crafted features and classical machine learning methods. Later, the focus shifted towards automated feature learning and end-to-end deep learning methods.

1.1 Traditional object detection approaches

Before the advent of deep learning, object detection relied on a combination of hand-crafted features and machine learning classifiers (Zou et al. 2023). Some of the notable traditional methods include:

- Correlation filters: Used to detect objects by correlating a filter with the image, such as matching templates (Park et al. 2019). These approaches struggle with variations in the appearance of objects and lighting conditions (Liu et al. 2021).
- Sliding window approach: This method involves moving a fixed-size window across the image and applying a classifier to each window to determine whether it contains an object (Teutsch and Kruger 2015). However, it struggles with varying object sizes and aspect ratios, which can lead to inaccurate detections and a high computational cost due to the exhaustive search involved.
- Viola–Jones detector: The Viola-Jones detector, introduced in 2001, uses Haar-like features (Lienhart and Maydt 2002) and a cascade of AdaBoost trained classifiers (Jun-Feng and Yu-Pin 2009) to detect objects in images efficiently (Li et al. 2012).

Supporting these methods are various hand-crafted feature extraction techniques, including:

- Gabor features: Extracted texture features using Gabor filters, which are effective for texture representation but can be computationally intensive (Hu et al. 2020).
- Histogram of oriented gradients (HOG): Captures edge or gradient structures that characterize the shape of objects, typically combined with Support Vector Machines (SVM) for classification (Surasak et al. 2018).
- Local binary patterns (LBP): Utilizes pixel intensity comparisons to form a binary pattern, used in texture classification and face recognition (Karis et al. 2016).
- Haar-like features: These features consider adjacent rectangular regions in a detection window, sum up the pixel intensities in each region, and calculate the difference between the sums. This difference is then used to categorize sub-regions in images (Mita et al. 2005).
- DPM (Deformable part models): DPM (Yan et al. 2014) *represents objects as a collection of deformable parts arranged in a spatial structure. It proved particularly effective for detecting objects under occlusions, pose variations, and cluttered backgrounds.*
- SIFT: Scale-Invariant Feature Transform (SIFT) is a robust method for detecting and describing local features in images (Piccinini et al. 2012). Beyond feature extraction, it has been effectively used for object detection by matching keypoints between input images and reference templates, leveraging its invariance to scale, rotation, and illumination changes.
- SURF: speeded-up robust features A faster alternative to SIFT, SURF detects and describes features using an efficient Hessian matrix approximation, making it suitable for real-time object detection tasks (Li and Zhang 2013).

1.1.1 Classification techniques used in traditional object detection methods

Some of the most commonly employed classification methods for these traditional object detectors include Support Vector Machine (SVM), statistical classifiers (e.g., Bayesian Classifier) and ensemble methods (e.g. Adaboost, Random Forest) and Multilayer Perceptrons (MLP) Neural Networks (Chiu et al. 2020; Mienye and Sun 2022).

These traditional methods in early computer vision systems, reliant on hand-crafted features and classical classifiers, offered moderate success under controlled conditions but struggled with robustness and generalization in diverse real-world scenarios, lacking the accuracy achieved by modern deep learning techniques (Xiang et al. 2014). Figure 1 shows

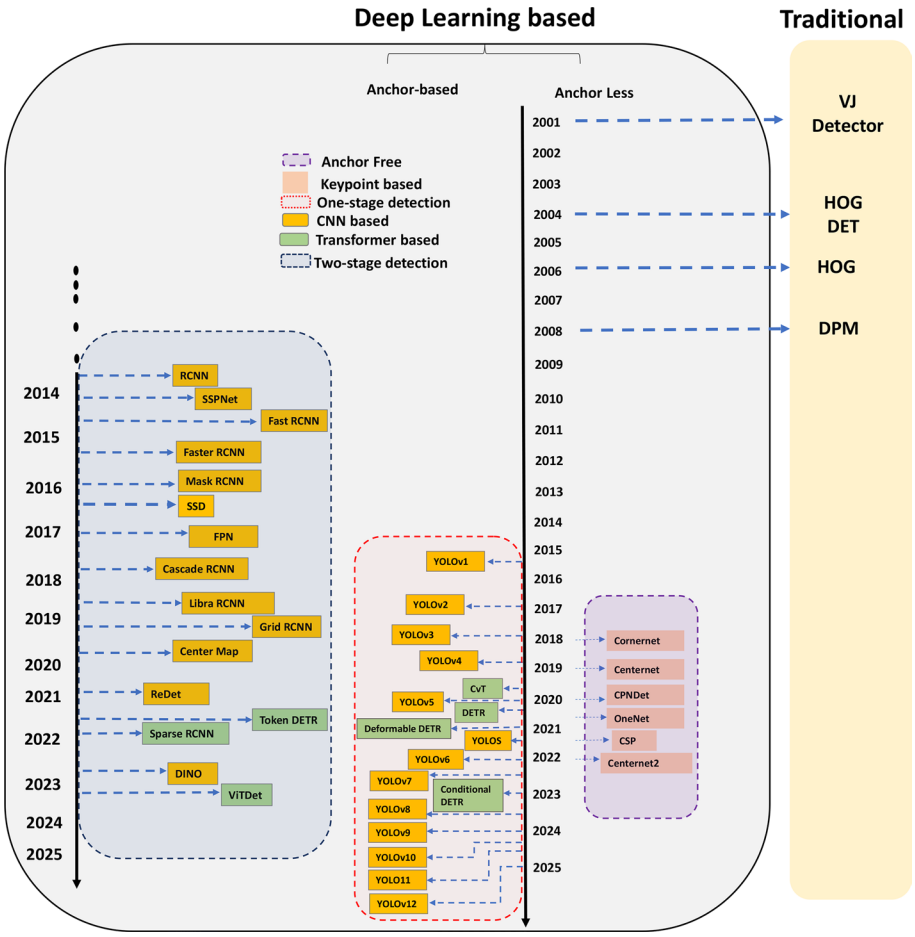


Fig. 1 Timeline of object detection paradigms and evolution of YOLO models. The figure shows the progression from traditional methods like VJ Detector and HOG to deep learning-based approaches, including R-CNN, Fast R-CNN, Faster R-CNN, and YOLO series. Recent advancements also highlight transformer-based models, such as DETR (Detection Transformer) (Carion et al. 2020) and ViTDet (Vision Transformer for Detection) (Li et al. 2022), which have demonstrated significant progress in object detection tasks

the historical development of computer vision systems emphasizing how object detection algorithms evolved.

1.2 Emergence of convolutional neural networks

After 2010, the performance of handcrafted features plateaued, leading to saturation in object detection research. However, in 2012, the world witnessed the emergence of convolutional neural networks (CNNs), marking a significant turning point in the field (Krizhevsky et al. 2012; Xie et al. 2021; Tang and Yuan 2015; Zhiqiang and Jun 2017). As a deep convolutional network, CNNs are able to learn robust and high-level feature representations of an image, and are particularly effective because:

- Hierarchical feature learning: CNNs learn to extract low-level features (e.g., edges, textures) in early layers and high-level features (e.g., object parts, shapes) in deeper layers, facilitating robust object representation (Li et al. 2020).
- Spatial invariance: Convolutional layers enable CNNs to recognize objects regardless of their position within the image, enhancing detection robustness (Crawford and Pineau 2019).
- Scalability and generalizability: CNNs can be scaled to handle larger datasets and more complex models, improving performance and robustness on a wide range of tasks and application environments (Tan et al. 2020).

However, CNNs can not be directly applied to the object detection task, due to varying numbers of objects, varying sizes, aspect ratios, and orientation. CNNs were primarily designed for image classification, meaning they output a single label for the entire image (Arkin et al. 2023). Whereas object detection tasks require not only classifying the object but also localizing it in the image, i.e., identifying the position of the object through bounding boxes.

1.3 Timeline of object detection paradigms and the evolution of YOLO models

One of the earliest deep learning-based object detectors was R-CNN, introduced in 2014 by Girshick et al. (2014). It marked a pivotal milestone in the development of detection models, breaking the stagnation in object detection by introducing Regions with CNN features (R-CNN). This groundbreaking approach revolutionized the field, sparking rapid advancements and accelerating the evolution of object detection at an unprecedented pace.

The idea behind R-CNN is simple, it uses the selective search algorithm to generate about 2000 region proposals, which are then processed by a CNN to extract features (Girshick et al. 2014). Finally, linear SVM classifiers are utilized to detect objects within each region and identify their respective categories. While R-CNN achieved significant progress, it has notable drawbacks: the redundant feature computations across a large number of overlapping proposals (over 2,000 boxes per image) result in extremely slow detection speeds, taking 14 s per image even with GPU acceleration (Xie et al. 2021).

After that, Fast R-CNN, introduced in 2015, improved object detection by addressing the redundant feature computations across numerous overlapping proposals (Bhat et al. 2023). It integrated region proposal feature extraction and classification into a single pass, significantly enhancing efficiency and speed compared to previous methods like R-CNN (Gir-

shick 2015). Building on this, Faster R-CNN advanced the approach further by introducing Region Proposal Networks (RPNs), enabling end-to-end training. This innovation eliminated the reliance on selective search, reducing computational complexity and streamlining the pipeline, thus allowing for faster and more accurate object detection without the need for external proposal generation (Ren et al. 2015; Mostafa et al. 2022).

Later, the Single Shot Multibox Detector (SSD) (Liu et al. 2016), introduced at the beginning of 2016, discretizes bounding boxes into predefined default boxes of various scales and aspect ratios. It predicts object scores and adjusts box shapes accordingly. By leveraging multi-scale feature maps, SSD handles objects of different sizes effectively. Unlike earlier methods, it eliminates the need for a separate proposal generation step, simplifying the training process and improving performance, particularly in detecting small objects.

These breakthroughs laid the foundation for numerous subsequent advancements in the field. Over the years, several other detectors have been developed, and by 2024, the YOLO series has become a dominant force in the domain, continuously evolving to improve speed, accuracy, and efficiency in real-time object detection tasks.

Fig. 1 presents a chronological overview of deep learning-based detectors, categorized into YOLO and others. Traditional detectors, shown on the right side, as well as Transformer-based detectors, are included for visual comparison.

1.4 You Only Look Once approach

The "You Only Look Once" (YOLO) object detection algorithm was first introduced by Redmon et al. (2016) in 2015, revolutionized real-time object detection by combining region proposal and classification into a single neural network, significantly reducing computation time. YOLO's unified architecture divides the image into a grid, predicting bounding boxes and class probabilities directly for each cell, enabling end-to-end learning (Redmon et al. 2016). YOLOv1 utilized a simplified CNN backbone, setting the stage with basic bounding box predictions. Subsequent versions like YOLOv2 incorporated the DarkNet-19 backbone and refined anchor boxes through K-means clustering. YOLOv3 expanded this further with a DarkNet-53 architecture, integrating multi-scale detection and residual connections. The series continued to innovate with YOLOv4 implementing CSPDarkNet-53 and PANet alongside mosaic data augmentation. YOLOv5 and YOLOv6 introduced CSPNet with dynamic anchor refinement and further enhancements in PANet, respectively. YOLOv7 featured an EfficientRep backbone with dynamic label assignment, while YOLOv8 introduced a Path Aggregation Network with Dynamic Kernel Attention. YOLOv9 developed multi-level auxiliary feature extraction, and YOLOv10 optimized the system with a lightweight classification head and distinct spatial and channel transformations. YOLOv11 introduced the C3k2 block in its backbone and utilized C2PSA for improved spatial attention. The latest, YOLOv12, marks a significant shift towards an attention-centric design, introducing the Area Attention (A^2) module for efficient large receptive field processing, Residual Efficient Layer Aggregation Networks (R-ELAN) for enhanced feature aggregation, and architectural optimizations including FlashAttention and adjusted MLP ratios. This attention-based approach allows YOLOv12 to achieve state-of-the-art performance in both accuracy and efficiency, surpassing previous CNN-based models while maintaining real-time detection capabilities (Tian et al. 2025). autonomous vehicles and traffic safety (Gheorghe et al. 2024; Wang et al. 2024; Shoman et al. 2024a, c), healthcare (Patel et al. 2023; Luo et al. 2021;

Salinas-Medina and Neme 2023), industrial applications (Pham et al. 2023; Klarák et al. 2024; Wang et al. 2024), surveillance and security (Arroyo et al. 2019; Bordoloi et al. 2020) and agriculture (Badgujar et al. 2024; Li et al. 2022; Fu et al. 2021; Zhong et al. 2018; Wang et al. 2022; Jiang et al. 2022; Chen et al. 2023; Yu et al. 2024; Jia et al. 2023; Umar et al. 2024; Sapkota et al. 2024a), where accuracy and speed are crucial.

In autonomous vehicles and traffic safety, YOLO has been widely adopted to improve safety in both road transport and aviation, allowing real-time object detection for collision avoidance, traffic monitoring, pedestrian detection, traffic sign detection, and aviation hazard analysis. This application aids in minimizing accidents and improving operational efficiency (Gheorghe et al. 2024; Bakirci and Bayraktar 2024; Wang et al. 2024).

In healthcare, YOLO has been instrumental in assisting and improving diagnostic processes and treatment outcomes. The applications include, but are not limited to, cancer detection (Prinzi et al. 2024; Aly et al. 2021), skin segmentation (Ünver and Ayan 2019), and pill identification (Tan et al. 2021; Suksawatchon et al. 2022) which showcase the model's ability to adapt to different needs, and essential tasks.

In industrial applications, YOLO aids in surface inspection processes to detect defects and anomalies (Pham et al. 2023; Klarák et al. 2024), structural health monitoring (Pratibha et al. 2023), reliability and safety (Fahim and Hasan 2024) ensuring quality control in manufacturing and production (Pham et al. 2023).

Surveillance and security systems also leverage YOLO for real-time monitoring and rapid identification of suspicious activities (Arroyo et al. 2019; Bordoloi et al. 2020). By integrating these models into surveillance systems, security personnel can more effectively monitor and respond to potential threats, enhancing public safety (Gorave et al. 2020). Similarly, in the context of public health measures like social distancing and face mask detection during pandemics (Kolpe et al. 2022; Bashir et al. 2023), YOLO models provided essential support in enforcing health regulations.

In agriculture, YOLO models have been applied to detect and classify crops (Ajayi et al. 2023), pests, and diseases (Morbekar et al. 2020; Li et al. 2022; Cheeti et al. 2021), facilitating precision agriculture techniques and automating farming operations to increase productivity and optimizing inputs. Additionally, in remote sensing, YOLO contributes to object recognition in satellite (Pham et al. 2020; Cheng et al. 2021) and aerial imagery (Chen et al. 2023; Luo et al. 2022), which supports urban planning, land use mapping, and environmental monitoring. These capabilities demonstrate YOLO's contribution to critical global challenges such as urban development and environmental conservation.

1.5 Motivation and organization of the study

Since "You Only Look Once" has been widely adopted in the field of computer vision, a search for this keyword in Google Scholar yields approximately 5,550,000 results as of June 9, 2024. The acronym "YOLO" further emphasizes its popularity, generating around 210,000 search results at the same time instant. Thousands of researchers have cited YOLO papers, highlighting its significant influence. This study aims to review and critically summarize the YOLO's decadal progress and its advancements over time, as visually summarized in the mind-map, shown in Fig. 2.

This comprehensive analysis starts with Sect. 2: *YOLO trajectory*, tracing the evolution from YOLOv1 to YOLOv12. In Sect. 3: *Context and distinctions of prior YOLO literature*

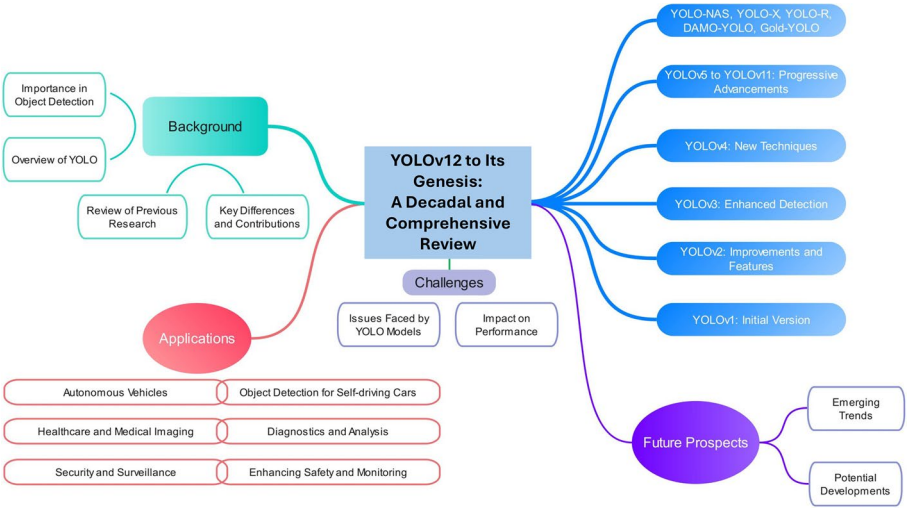


Fig. 2 A schematic of the manuscript. We discuss all versions of YOLO, with comparative analysis. Applications in key areas; such as, autonomous vehicles, and security and surveillance are also presented. Challenges in each YOLO version, with performance enhancements, are also highlighted, we also provide visionary thoughts on the future impact of YOLO on industry and society

offers insights into the background and unique aspects of earlier studies. Section 4: *Review of YOLO versions* details the key features and improvements of each version. In Sect. 5: *Applications* various use cases across different domains are highlighted, demonstrating the versatility of YOLO models. Following this, Sect. 6 *Challenges, limitations and future directions* addresses current issues and potential advancements. Finally, the *Conclusion* section summarizes the findings of this comprehensive review. Each section is further divided into various sub-subsections to present and discuss specific topic areas relevant to the corresponding sections.

2 The evolution of YOLO: trajectory and variants

Fig. 1 illustrates the development timeline of the YOLO models, beginning with the release of YOLOv1 and progressing to the latest version, YOLO12. This timeline highlights the key advancements and iterations in the YOLO series.

YOLOv1 (Redmon et al. 2016) was introduced in 2016 as a novel approach to object detection, offering good accuracy and computational speed by processing images using a single stage network architecture. The first YOLO version laid the foundation for real-time applications of machine vision systems, setting a new standard for subsequent developments.

YOLOv2, or YOLO9000 (Li and Yang 2018; Nakahara et al. 2018), expanded on the foundation of YOLOv1 by improving the resolution at which the model operated and by expanding the capability to detect over 9000 object categories, thus enhancing its versatility and accuracy. YOLOv2 introduced two primary variants: a smaller version optimized for speed and a larger version focused on higher accuracy.

YOLOv3 further advanced these capabilities by implementing multi-scale predictions and a deeper network architecture, which allowed better detection of smaller objects (Kim et al. 2018). YOLOv3 introduced three primary variants, each designed to balance model size and performance: YOLOv3-spp (Small), Standard, and YOLOv3-tiny(Tiny), catering to different trade-offs between speed and accuracy.

The series continued to evolve with YOLOv4 and YOLOv5, each introducing more refined techniques and optimizations to improve detection performance (i.e., accuracy and speed) even further (Nepal and Eslamiat 2022; Sozzi et al. 2022; Mohod et al. 2022). YOLOv4 introduced four main variants: the standard version, YOLOv4-CSP, which incorporates Cross-Stage Partial (CSP) networks to enhance performance and reduce computational cost; YOLOv4x-mish, which utilizes the Mish activation function to improve accuracy while maintaining efficiency; and YOLOv4-tiny, a lightweight version optimized for real-time applications and edge devices, sacrificing some accuracy for speed. YOLOv5, developed by Ultralytics (2020) [1], brought significant improvements in terms of ease of use and performance, establishing itself as a popular choice in the computer vision community. YOLOv5 introduced five primary variants to meet various performance needs: YOLOv5s (small), optimized for speed and efficiency in resource-constrained environments; YOLOv5m (medium), offering a balanced trade-off between speed and accuracy; YOLOv5l (large), designed for higher accuracy at the expense of resources; YOLOv5x (extra-large), focused on top-tier accuracy for powerful hardware; and YOLOv5n (nano), a lightweight version tailored for rapid inference and low computational demands, ideal for real-time applications and edge devices.

Subsequent versions, YOLOv6 through YOLO11, have continued to build on this success, focusing on enhancing model scalability, reducing computational demands, and improving real-time performance metrics.

Li et al. (2022) introduced YOLOv6 in 2022. Developed by a team from Meituan, a Chinese e-commerce platform, YOLOv6 features a novel backbone and neck architecture. It also incorporates advanced training techniques such as Anchor-Aided Training (AAT) and Self-Distillation to enhance performance and efficiency. YOLOv6 introduces three main variants: the standard version, balancing accuracy and speed for general detection tasks; YOLOv6-Nano, optimized for real-time applications with a focus on speed and performance on edge devices; and YOLOv6-Tiny, designed for even faster inference on low-resource hardware, trading off some accuracy.

YOLOv7 (Wang et al. 2022, 2023) introduces advanced techniques like trainable bag-of-freebies (optimizations that improve accuracy without increasing inference cost) and dynamic label assignment. It introduces three variants: the standard version, balancing speed and accuracy; YOLOv7-X, a more powerful variant optimized for performance but requiring more computational resources; and YOLOv7-Tiny, a lightweight version designed for real-time applications on edge devices, prioritizing speed over accuracy.

YOLOv8 was released in 2023 by Ultralytics (Jocher et al. 2023). It features a more efficient architecture, enhanced training techniques, and support for larger datasets. Its user-friendly implementation in PyTorch makes it accessible for both research and production. YOLOv8 introduces four variants: YOLOv8-S, optimized for fast inference on edge devices with some accuracy trade-offs; YOLOv8-M, balancing accuracy and speed for general tasks;

[1] Ultralytics specializes in AI and deep learning, known for YOLOv5, YOLOv8, and YOLO11 models used in object detection, segmentation, and classification for computer vision tasks.

YOLOv8-L, prioritizing accuracy at the cost of computational demand; and YOLOv8-Tiny, a lightweight version for real-time applications.

YOLOv9 (Wang et al. 2024) proposed the concept of programmable gradient information (PGI) to cope with the various changes required by deep networks to achieve multiple objectives. PGI can provide complete input information for the target task to calculate objective function, so that reliable gradient information can be obtained to update network weights. In addition, a new lightweight network architecture – Generalized Efficient Layer Aggregation Network (GELAN), based on gradient path planning is designed. GELAN's architecture confirms that PGI has gained superior results on lightweight models. We verified the proposed GELAN and PGI on MS COCO dataset based object detection. Its variants are YOLOv9t, YOLOv9s, YOLOv9m, YOLOv9c, YOLOv9e (Ultralytics 2023a).

YOLOv10 (Ultralytics 2023b), developed by researchers at Tsinghua University, introduces a novel approach to real-time object detection, addressing the limitations of both post-processing and model architecture in previous YOLO versions. By eliminating non-maximum suppression (NMS) and optimizing key components of the model, YOLOv10 offers significant improvements in efficiency and performance. This version introduces six distinct variants as YOLOv10-N, YOLOv10-S, YOLOv10-M, YOLOv10-B, YOLOv10-L, and YOLOv10-X (Wang et al. 2024).

Notably, YOLOv10-N and YOLOv10-S exhibit the lowest latencies at 1.84 ms and 2.49 ms, respectively, making them highly suitable for applications requiring low latency. These models outperform their predecessors, with YOLOv10-X achieving the highest mAP of 54.4% and a latency of 10.70 ms, reflecting a well-balanced enhancement in both accuracy and inference speed.

YOLOv11 represents a significant advancement in object detection, featuring a sophisticated backbone and neck architecture for enhanced feature extraction. It optimizes speed and efficiency while maintaining high accuracy. YOLOv11 balances precision and computational efficiency, suitable for various applications from embedded systems to large-scale deployments. The model comes in five variants: YOLOv11n, YOLOv11s, YOLOv11m, YOLOv11L, and YOLOv11x, based on network depth.

YOLOv12 (Tian et al. 2025) further revolutionizes the field with an attention-centric approach. It introduces the Area Attention (A^2) module and Residual Efficient Layer Aggregation Networks (R-ELAN) for improved feature processing. Utilizing these architectural changes, YOLOv12 achieves state-of-the-art performance while maintaining real-time detection capabilities. For instance, YOLOv12-N was reported to achieve 40.6% mAP with a 1.64 ms inference latency on a T4 GPU, outperforming YOLOv10-N and YOLOv11-N by 2.1 mAP with a comparable speed. YOLOv12 also demonstrated superior object contour definition and foreground activation compared to its predecessors. It supports various tasks, including object detection, segmentation, classification, pose estimation, and oriented object detection, making it a versatile tool for diverse computer vision applications. YOLOv12 is available in five variants: YOLOv12-N, YOLOv12-S, YOLOv12-M, YOLOv12-L, and YOLOv12-X, each offering different trade-offs between performance and computational requirements (Tian et al. 2025).

To summarize, each iteration of the YOLO series has set new benchmarks for object detection capabilities and significantly impacted various application areas, from autonomous vehicles and traffic safety

2.1 Significance of latency and mAP scores in YOLO

Inference Time (T_{inf}) and mean Average Precision (mAP) are critical metrics to assess the performance of object detection models such as YOLO (Tang and Yuan 2015; Mao et al. 2019). Inference Time specifically measures the duration required for the model to process an image and generate predictions, focusing solely on the computational phase and is typically measured in milliseconds (ms) (Mao et al. 2019). This metric excludes any delays from image pre-processing or post-processing, providing a clear measure of the model's computational efficiency. Lower inference times are crucial for real-time applications such as autonomous driving, surveillance, and robotics, where rapid and accurate detections are essential (Chen et al. 2020). High inference times can lead to delays that compromise safety and effectiveness in these dynamic settings (Pestana et al. 2021).

Frames Per Second (FPS) is another essential metric that indicates how many images the model can evaluate each second, complementing inference time by illustrating the model's ability to handle streaming video or rapid image sequences. Both inference time and FPS provide a detailed view of the real-time operational performance of a model.

It is also important to note that these performance metrics are highly dependent on the hardware platform used for testing. Differences in computational power can significantly influence results, which makes it essential to standardize hardware during benchmark tests to ensure fair comparisons. Likewise, mAP is a comprehensive metric used to evaluate the accuracy of object detection models (Zhou et al. 2018). It considers both precision and recall (Table 1), and it is calculated by taking the average precision (AP) across all classes and then averaging these AP scores (Hall et al. 2020; Zhou et al. 2018). It provides a balanced view of how well the model performs across different object categories and varying conditions within the dataset. Other metrics used for comprehensive evaluation of YOLO models (Goutte and Gaussier 2005; Liang et al. 2021) are detailed in Table 1.

Table 1 Summary of performance metrics used in model evaluation

No.	Performance metric	Symbol	Equation	Description
1	Precision	P	$P = \frac{TP}{TP+FP}$	Ratio of true positive detections to the total predicted positives
2	Recall	R	$R = \frac{TP}{TP+FN}$	Ratio of true positive detections to the total actual positives
3	F1 score	$F1$	$F1 = 2 \cdot \frac{P \cdot R}{P+R}$	Harmonic mean of precision and recall, balancing both metrics to provide a single performance measure for the model
4	Intersection over union	IoU	$\text{IoU} = \frac{\text{Area of overlap}}{\text{Area of union}}$	Measures the overlap between the predicted and actual bounding boxes
5	Frames per second	FPS	$\text{FPS} = \frac{1}{L}$	Number of images the model processes per second, inversely related to latency
6	Non-maximum suppression	NMS	–	NMS is a post-processing step in YOLO to remove redundant bounding-boxes

Here, True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN) are the key performance evaluators. TP is instance where the model correctly identifies an object as present. TN occurs when the model correctly predicts the absence of an object. FP arises when the model incorrectly identifies an object as present, and FN happens when the model fails to detect an object that is actually present. These metrics are crucial for assessing the accuracy and reliability of the YOLO object detection (Hall et al. 2020; Zhou et al. 2018; Liang et al. 2021).

2.2 Single-stage detection with YOLO

The Single Shot MultiBox Detector (SSD) (Liu et al. 2016) (Fig. 3) introduced in 2016 revolutionized object detection by streamlining the process through a single-stage approach, significantly inspiring subsequent developments in YOLO models (Liu et al. 2016; Fu et al. 2017; Zhang et al. 2018). Unlike two-stage models like R-CNN, which rely on a region proposal step before actual object detection, SSD and by extension, YOLO variants, perform detection and classification in a single sweep across the image. This paradigm shift enhances the detection process by eliminating intermediate steps, thus facilitating faster and more efficient object detection suitable for real-time applications. The architecture of SSD, which YOLO models have adapted, utilizes multiple feature maps at different resolutions to detect objects of various sizes, employing a diverse array of anchor boxes at each feature map location to improve localization accuracy (Cui et al. 2018; Lin et al. 2017).

Figure 3 illustrates a YOLO model that incorporates SSD's architectural principles to enhance real-time detection capabilities through improved feature extraction using Multi-Headed Attention layers. This adoption from SSD methodology significantly boosts the processing speed and detection accuracy of models such as YOLOv8, YOLOv9, and YOLOv10,

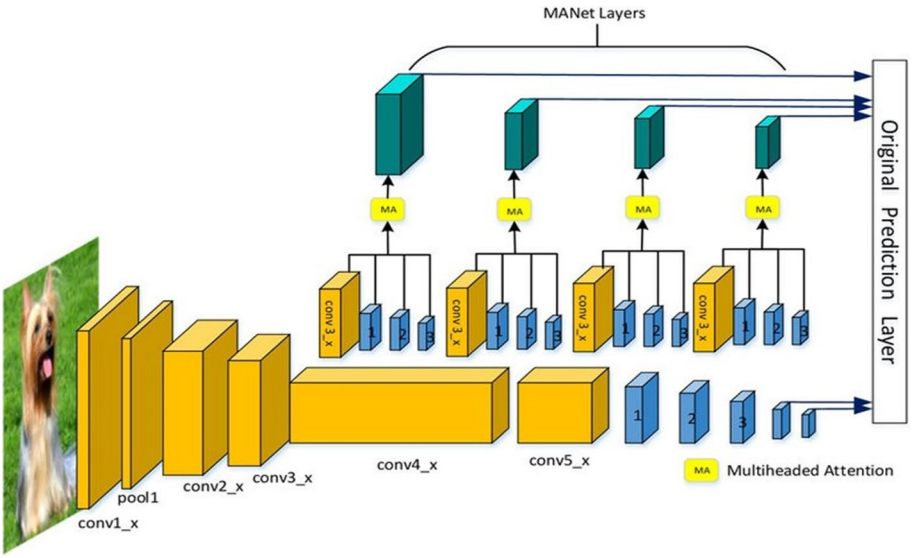


Fig. 3 Enhanced YOLO model architecture incorporating SSD's single-stage detection approach with Multi-Headed Attention (MA) layers for superior real-time object detection performance (Jiang et al. 2019)

making them ideal for rapid and reliable object detection in resource-constrained environments (Tang et al. 2018; Li et al. 2017). The efficient single-shot mechanism, which directly classifies and localizes objects, highlights the ongoing evolution of the YOLO series to meet the accuracy and speed requirements of diverse real-world scenarios (Zhang et al. 2018).

3 Prior YOLO literature: context and distinctions

We collected the existing published literature on YOLO to document and critically analyze the past knowledge, including major highlights and limitations, which are briefly summarized and discussed here:

- *A review of YOLO algorithm developments* by Jiang et al. (2022) provided an insightful overview of YOLO algorithm development and its evolution through its versions. The authors analyze the fundamental aspects of YOLO's to object detection, comparing its various iterations to traditional CNNs. They emphasize ongoing improvements in YOLO, particularly in enhancing target recognition and feature extraction capabilities. It also discusses the application of YOLO in specific fields, such as finance, highlighting its practical implications in feature extraction for image-based news analysis (Jiang et al. 2022).
- *A comprehensive systematic review of YOLO for medical object detection (2018 to 2023)* by Ragab et al. (2024) presented a systematic review of YOLO's application in the medical field, that analyzes how different variants, particularly YOLOv7 and YOLOv8, have been employed for various medical detection tasks. They highlight the algorithm's significant performance in lesion detection, skin lesion classification, and other critical areas, demonstrating YOLO's superiority over traditional methods in accuracy and computational efficiency. Despite its successes, the review identifies challenges, such as the need for well-annotated datasets and addresses the high computational demands of YOLO implementations. The paper suggested directions for future research to optimize YOLO's application in medical object detection (Ragab et al. 2024).
- *A comprehensive review of YOLO architectures in computer vision: from YOLOv1 to YOLOv8 and YOLO-NAS* by Terven et al. (2023) provides an extensive analysis of the evolutionary trajectory of the YOLO algorithm, detailing how each iteration has contributed to advances in real-time object detection. Their review covers the significant architectural and training enhancements from YOLOv1 through YOLOv8 and introduces YOLO-NAS and YOLO with Transformers. This study serves as a valuable resource for understanding the progression in network architecture, which has progressively improved YOLO's efficacy in diverse applications such as robotics and autonomous driving.
- *YOLOv1 to v8: unveiling each variant-a comprehensive review of YOLO* by Hussain (2024), provided in-depth analyses of the internal components and architectural innovations of each YOLO variant. It provided a deep dive into the structural details and incremental improvements that have marked the evolution of YOLO, presenting a well-structured analysis complete with performance benchmarks. This methodological approach not only highlights the capabilities of each variant but also discusses their practical impact across different domains, suggesting the potential for future enhancements

like federated learning to improve privacy and model generalization (Hussain 2024).

- *YOLO-v1 to YOLO-v8, the rise of YOLO and its complementary nature toward digital manufacturing and industrial defect detection* by Hussain (2023) reviewed and showed rapid progression of the YOLO variants, focusing on their critical role in industrial applications, specifically for defect detection in manufacturing. Starting with YOLOv1 and extending through YOLOv8, the paper illustrates how each version has been optimized to meet the demanding needs of real-time, high-accuracy defect detection on constrained devices. Hussain's work not only examines the technical advancements within each YOLO iteration but also validates their practical efficacy through deployment scenarios in the manufacturing sector, emphasizing YOLO's alignment with industrial needs (Hussain 2023).
- *YOLOv1 to YOLOv10: The fastest and most accurate real-time object detection systems* by Wang et al. (2024) provide a comprehensive literature survey on the YOLO series, from YOLOv1 to YOLOv10. This review uniquely revisits the characteristics of YOLO through a contemporary technical lens, highlighting its ongoing influence in advancing real-time computer vision research and subsequent technological developments. The authors explore the evolution of YOLO's methodologies over the past decade and its diverse applications in fields requiring real-time object analysis. By doing so, they underscore YOLO's role as a foundational technology in various computer vision tasks, such as instance segmentation and 3D object detection.
- *Evaluating the evolution of YOLO models: a comprehensive benchmark study of YOLO11 and its predecessors* by Jegham et al. (2024) conducts a detailed benchmark analysis of YOLO models from YOLOv3 to YOLO11. It examines their performance across three diverse datasets: Traffic Signs, African Wildlife, and Ships and Vessels, focusing on different challenges like object size and aspect ratio. Employing metrics like Precision, Recall, mAP, Processing Time, GFLOPs, and Model Size, the study identifies the strengths and limitations of each version, with YOLO11m showing exceptional balance in accuracy and efficiency across the datasets.

For the comprehensive review articles, it is advantageous to pinpoint a specific gap that the proposed review will address. For instance, a common oversight in the existing literature is the omission of the latest YOLO iterations, particularly YOLOv9, YOLOv10, YOLOv11 and YOLOv12 or neglecting to cover the application domains of interest. Given the YOLO algorithm's ten-year milestone, there is a pressing need to systematically document and critically evaluate these newer models. Our review aims to fill this void by providing updated, in-depth insights and comparative analysis of YOLOv9 and YOLOv12, extending across various applications to serve the wider research and technical community. This state-of-the-art review intends to highlight the continued advancements and capabilities of these models within the dynamic field of object detection technology.

In this review paper, we adopt a unique reverse-chronological approach to analyze the progression of YOLO, beginning with the most recent versions and moving backwards. The analysis is divided into six distinct subsections. The first subsection covers the latest iterations, YOLOv12 and YOLOv11, The second subsection examines YOLOv10, YOLOv9, and YOLOv8, where we delve into the architecture and advancements that define the forefront of object detection technology. This approach not only shows the most cutting-edge developments but also sets the stage for understanding the incremental improvements

that have been realized over time. The third subsection reviews YOLOv7, YOLOv6, and YOLOv5, tracing further back in the series to highlight the evolutionary steps contributing to the enhancements observed in the later versions. We analyze each model's technical and scientific aspects to provide a comprehensive view of the progress within these iterations. The fourth subsection addresses the earlier YOLO versions, offering a complete historical perspective that enriches the reader's understanding of the foundational technologies and the methodologies, refined through successive updates. The fifth subsection presents alternative versions derived from YOLO.

To close this section, we discuss the application of the YOLO models in reverse order across five critical real-world domains: autonomous vehicles and traffic safety, healthcare and medical image analysis, surveillance and security, industrial manufacturing and agriculture. For each application, we present a detailed examination and corresponding tabular data in reverse chronological order, showcasing how YOLO technologies have been adapted and implemented to meet specific industry needs and challenges. This reverse review strategy not only emphasizes the state-of-the-art but also provides a narrative of technological evolution, illustrating how each iteration builds upon the last to push the boundaries of what's possible in object detection. By understanding where YOLO technology stands today and how it got there, readers gain a comprehensive view of its capabilities and potential future directions. This methodical unpacking of the YOLO series not only highlights technological advancements but also offers insights into the broader implications and utility of these models in practical scenarios, setting the groundwork for anticipating future innovations in object detection technology.

4 Review of YOLO versions

This section reviews YOLO series models, starting from the advanced and latest version, YOLOv12, and progressively tracing back to the foundational YOLOv1. By first highlighting the most recent technological advancements, this approach enables immediate insights into the state-of-the-art capabilities of object detection. Subsequently, the narrative explores how earlier models laid the groundwork for these innovations.

4.1 YOLOv12 and YOLO11

YOLOv12 (Tian et al. 2025) is the most recent YOLO version introduced in February 2025, which marks a substantial advancement in real-time object detection by integrating attention mechanisms into the YOLO framework while maintaining competitive inference speeds. This attention-centric framework not only surpasses popular real-time detectors (e.g., Faster R-CNN, RetinaNet, Detectron 2) in accuracy but also achieves state-of-the-art performance through a combination of innovative attention methods, Residual Efficient Layer Aggregation Networks (R-ELAN), and several architectural optimizations.

Fig. 4a demonstrates latency comparisons on the MS COCO benchmark dataset, highlighting YOLOv12's significantly lower inference latency compared to YOLOv11, YOLOv10, YOLOv9, and YOLOv8. The curve reveals a substantial reduction in latency, enabling faster processing speeds while maintaining high detection accuracy (Tian et al. 2025). Complementing this, Fig. 4b presents comparison in terms of GFLOPs, which shows

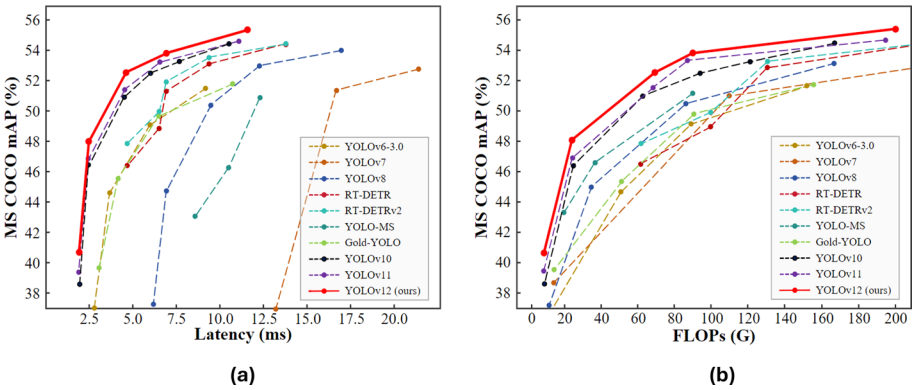


Fig. 4 **a** Latency comparison on the MS COCO benchmark reveals significantly faster inference achieved with YOLOv12 compared to the same achieved with previous YOLO versions. **b** GFLOPs analysis also shows enhanced computational efficiency. (Tian et al. 2025)

YOLOv12 achieves higher computational efficiency, reflecting its ability to handle complex computations effectively. This balance between speed and computational power demonstrates YOLOv12's robust performance.

On the COCO dataset, YOLOv12 sets a new state-of-the-art standard, with the lightweight YOLOv12-N achieving 40.6% mAP and the larger YOLOv12-X reaching 55.2% mAP. These results, combined with the latency and GFLOPs improvements, establish YOLOv12 as a new benchmark in real-time object detection. The model framework is available in five scales: YOLOv12-N, S, M, L, and X, each optimized for specific applications. For instance, YOLOv12-N achieves 40.6% mAP at 1.64 ms on a T4 GPU, outperforming YOLOv10-N and YOLOv11-N by 2.1% and 1.2% mAP, respectively. Similarly, YOLOv12-S attains 48.0% mAP at 2.61 ms/image, surpassing YOLOv8-S, YOLOv9-S, YOLOv10-S, and YOLOv11-S by margins of 3.0%, 1.2%, 1.7%, and 1.1% mAP. The larger models in the YOLOv12 family continue to show improvements in efficiency and performance. Notably, YOLOv12-M achieves 52.5% mAP at 4.86 ms/image. In terms of computational effieciency, YOLOv12-L demonstrates a significant reduction in FLOPs, decreasing by 31.4G compared to its predecessor, YOLOv10-L. At the highest end of the scale, YOLOv12-X showcases superior performance, outperforming both YOLOv10-X and YOLOv11-X in detection accuracy (Tian et al. 2025).

YOLOv12 also surpasses end-to-end detectors like RT-DETR and RT-DETRv2. For example, YOLOv12-S runs 42% faster than RT-DETR-R18 and RT-DETRv2-R18, using only 36% of the computation and 45% of the parameters. Residual connections show minimal impact on convergence in smaller models (YOLOv12-N) but are critical for stable training in larger models (YOLOv12-L/X), with YOLOv12-X requiring a scaling factor of 0.01. The area attention module reduces inference time by 0.7 ms on an RTX 3080 with FP32 precision, while FlashAttention further accelerates inference by 0.3-−0.4 ms.

Visualization analyses confirm that YOLOv12 produces clearer object contours and more precise foreground activations than its predecessors. A convolution-based attention implementation proves to be faster than linear alternatives. Additionally, a hierarchical design, extended training (approximately 600 epochs), an optimized convolution kernel size

(7×7), absence of positional embedding, and an MLP ratio of 1.2 collectively enhance the framework's performance and efficiency.

As discussed before, the YOLOv12 architecture (Fig. 5a), demonstrates an advanced integration of A^2 (Area Attention) modules, R-ELAN (Residual Efficient Layer Aggregation Networks) blocks, and a streamlined detection head. This design optimizes the model's visual information processing while maintaining high accuracy. The major innovations on the YOLOv12 architecture are listed below.

4.1.1 YOLOv12 architectural innovation

- Area attention (A^2) module: This module implements segmented feature processing with Flash Attention integration, reducing computational complexity by 50% through spatial reshaping while maintaining large receptive fields. AA enables real-time detection at fixed $n = 640$ resolution through optimized memory access patterns, as illustrated in Fig. 5a.
- Residual ELAN (R-ELAN) hierarchy: R-ELAN combines residual shortcuts (0.01 scaling) with dual-branch processing to mitigate the gradient vanishing problem. The model also features a streamlined final aggregation stage that reduces parameters by 18% and FLOPs by 24% compared to baseline architectures, as shown in Fig. 5b.

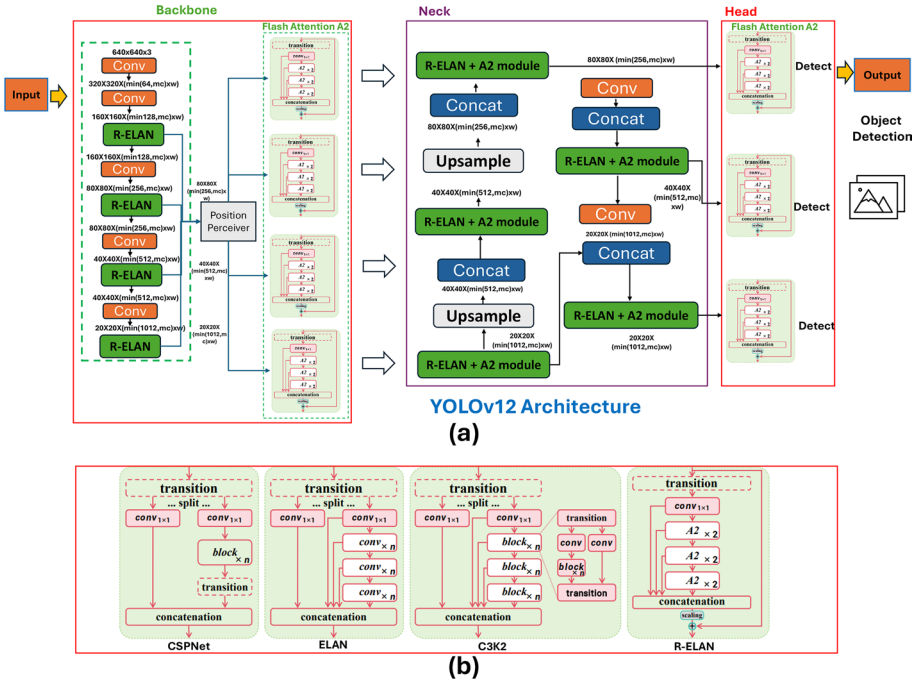


Fig. 5 **a** Architecture of the YOLOv12 object detection model that integrated Area Attention (A^2) modules, R-ELAN blocks, and a streamlined detection head; **b** Comparison of "Attention Module" architectures: CSPNet (Wang et al. 2020), ELAN (Wang et al. 2022), C3K2 (used in YOLOv9) (Wang et al. 2024), and the novel R-ELAN (Tian et al. 2025) introduced with YOLOv12, which improved residual connections and enhanced feature aggregation, demonstrating superior performance

- Efficient architectural revisions: YOLOv12 replaces positional encoding with 7×7 depth-wise convolution for implicit spatial awareness. It also implements adaptive MLP ratio (1.2×) and shallow block stacking to balance the computational load, achieving 4.1 ms inference latency on V100 hardware.
- Optimized training framework: The model was trained over 600 epochs using SGD with cosine scheduling (initial lr = 0.01). The model also incorporates Mosaic-9 and Mixup augmentations with 12.8% mAP gain on COCO dataset, maintaining real-time performance through selective kernel convolution integration.

Figure 5b presents an architectural comparison of popular attention modules: CSPNet, ELAN, C3K2 (a case of GELAN), and the proposed R-ELAN. Brief summary of these modules is blow.

- CSPNet (Cross stage partial network): CSPNet enhances gradient flow by splitting feature maps into two paths, one for learning and one for propagation, reducing computational bottlenecks and improving inference speed. This model is visually depicted in Fig. 5b (leftmost module).
- ELAN (Efficient layer aggregation network): ELAN improves feature integration by aggregating multi-scale features efficiently, enhancing the model's ability to detect objects at various scales. However, as shown in Fig. 5b (second module), ELAN can introduce instability due to gradient blocking and lacks of residual connections, particularly in large-scale models.
- C3K2 (Compact GELAN): This module is a compact version of GELAN (Generalized Efficient Layer Aggregation Network) that offers a balance between computational efficiency and feature expressiveness, suitable for resource-constrained environments. The module is also illustrated in Fig. 5b (third module).
- R-ELAN (Residual ELAN): R-ELAN introduces residual connections and redesigns feature aggregation to address optimization challenges in attention-based models, combining the benefits of residual learning with efficient feature aggregation. As shown in Fig. 5b (rightmost module), R-ELAN applies a residual shortcut with a scaling factor (default 0.01) and processes the input through a transition layer, followed by a bottleneck structure for improved stability and performance.

The R-ELAN design, as depicted in Fig. 5b, addresses the limitations of ELAN by introducing residual connections and a revised aggregation approach. Unlike ELAN, which splits the input into two parts and processes them separately, R-ELAN applies a transition layer to adjust channel dimensions and processes the feature map through subsequent blocks before concatenation. This design mitigates gradient blocking and ensures stable convergence, particularly in large-scale models like YOLOv12-L and YOLOv12-X. The integration of residual connections and attention mechanisms in R-ELAN, as shown in Fig. 5b, highlights YOLOv12's architectural advancements in balancing efficiency and accuracy.

YOLO11: YOLO11 developed by Ultralytics represents the most recent version building upon the foundations established by its predecessors in the YOLO family. This latest iteration introduces several architectural innovations that enhance its performance across a wide spectrum of tasks as depicted in Fig. 6. The model incorporates the C3k2 (Cross Stage Partial with kernel size 2) block, which replaces the C2f block used in previous ver-

sions, offering improved computational efficiency (Sapkota et al. 2024b). Additionally, YOLOv11 retains the SPPF (Spatial Pyramid Pooling-Fast) component and introduces the C2PSA (Convolutional block with Parallel Spatial Attention) module, collectively enhancing feature extraction capabilities (Sapkota et al. 2024a). These architectural enhancements enable YOLOv11 to capture intricate image details with greater precision, particularly in challenging scenarios involving small or occluded objects. The model's versatility is evident in its support for a broad range of computer vision tasks, including object detection, instance segmentation, pose estimation, image classification, and oriented bounding box (OBB) detection (Ultralytics 2024).

Empirical evaluations of YOLOv achieves a higher mean Average Precision (mAP) score on the COCO dataset while utilizing 22% fewer parameters compared to its YOLOv8m counterpart (Jegham et al. 2024). This reduction in parameter count contributes to faster model performance without significantly impacting overall accuracy. Furthermore, YOLOv11 exhibits inference times approximately 2% faster than YOLOv10, making it particularly well-suited for real-time applications. The model's efficiency extends across various deployment environments, including edge devices, cloud platforms, and systems supporting NVIDIA GPUs. YOLOv11 is available in multiple variants, ranging from nano to extra-large, catering to diverse computational requirements and use cases. These advancements position YOLOv11 as a state-of-the-art solution for industries requiring rapid and accurate image analysis, such as autonomous driving, surveillance, and industrial automation.

YOLOv10, incorporates advanced techniques like automated architecture search and more refined loss functions to enhance detection accuracy and speed, tailored for both edge and cloud computing environments. This version and its predecessors, YOLOv9 and YOLOv8, introduce substantial improvements in network architecture, such as the integration of cross-stage partial networks (CSPNets) and the use of transformer-based backbones for better feature extraction across different scales. YOLOv7 and YOLOv6 continued to build on these improvements by optimizing computational efficiency and expanding model scalability. Meanwhile, YOLOv5 introduced PyTorch support, which significantly enhanced the model's accessibility and adaptability, thus broadening its application in industry and academia. YOLOv4, on the other hand, marked a pivotal point in YOLO history by integrating

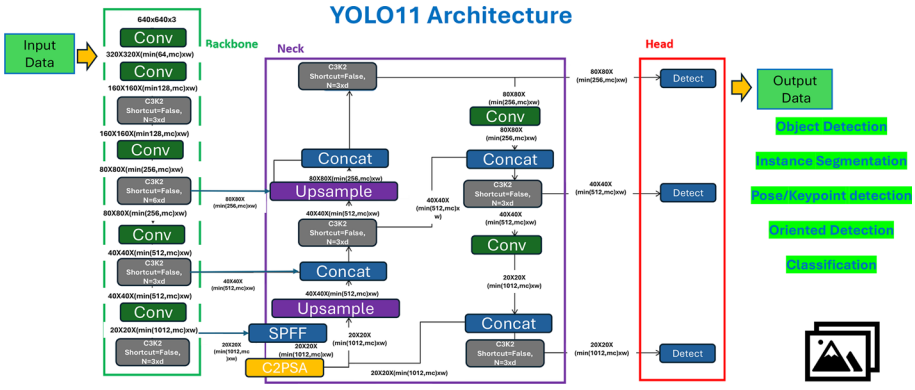


Fig. 6 YOLOv11 architecture diagram: enhanced backbone with C3k2 blocks replacing C2f, SPPF for multi-scale feature extraction, C2PSA for attention mechanism, and optimized neck. Efficient feature processing through multiple scales, culminating in a detect head for multi-class object detection and localization

features like Mish activation and Cross-Stage Partial connections, setting new standards for speed and accuracy in real-time applications. The mid-generations, starting with YOLOv3, were notable for introducing multi-scale predictions and bounding box predictions across different layers, which greatly improved the model's ability to detect small objects-a long-standing challenge in earlier versions. YOLOv3 was also one of the first YOLO models to leverage deeper feature extractors like Darknet-53, which significantly boosted its performance over YOLOv2. YOLOv2 itself had introduced important features such as batch normalization and high-resolution classifiers, which enhanced the overall accuracy without compromising the speed. The original YOLO model, YOLOv1, was revolutionary, proposing a single-stage detection framework that unified the object detection process into a single neural network model.

Fig. 7a illustrates a sophisticated transformer-based model that simplifies the detection process by integrating dual label assignments and eliminating the need for non-max suppression (NMS), achieving a streamlined, end-to-end object detection. YOLOv9, as shown in Fig. 7b, introduces the Programmable Gradient Information (PGI) system to enhance model interpretability and robustness, significantly improving generalization across various tasks. Moving to YOLOv8 and YOLOv7, Figs. 7c and d respectively depict their architectures which incorporate elements like ELAN and CSPNet to boost performance and flexibility across computing devices. YOLOv6, highlighted in Fig. 7e, focuses on industry applications with enhancements in model quantization and real-time performance.

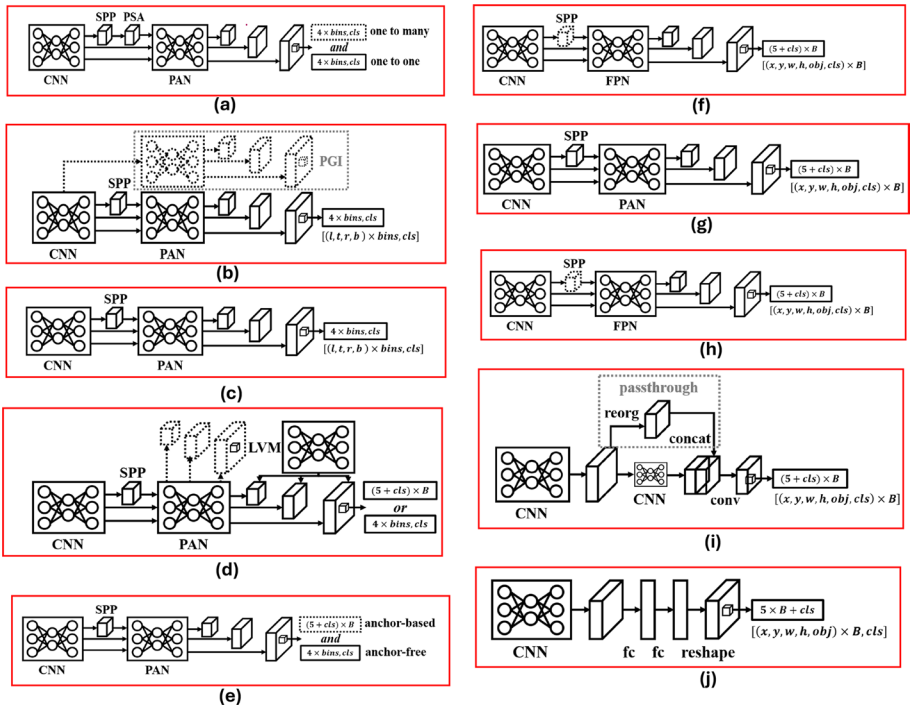


Fig. 7 Simplified Architecture diagrams for: **a** YOLOv10; **b** YOLOv9; **c** YOLOv8; **d** YOLOv7; **e** YOLOv6; **f** YOLOv5; **g** YOLOv4; **h** YOLOv3; **i** YOLOv2; **j** YOLOv1. The diagrams are detailed in Wang et al. (2024)

YOLOv5, represented in Fig. 7f, marks a pivotal development with its adoption of PyTorch and improvements in training methods that enhance model accessibility and efficiency. In contrast, YOLOv4, shown in Fig. 7g, integrates technologies like CSPNet and Path Aggregation Network (PAN) (Liu et al. 2018) to optimize real-time detection. YOLOv3, visualized in Fig. 7h, introduces significant architectural changes with Darknet-53 and multi-scale predictions, which substantially enhance the detection of small objects. YOLOv2, depicted in Fig. 7i, advances the architecture with dimension clusters and fine-grained features, improving the model's efficiency and adaptability. Finally, YOLOv1, as outlined in Fig. 7j, revolutionizes object detection by integrating a single-stage detector that performs grid-based predictions in real-time, significantly reducing model complexity and enhancing speed.

4.2 YOLOv10, YOLOv9 and YOLOv8

YOLOv10 (Wang et al. 2024), developed at Tsinghua University, China, represents a breakthrough in the YOLO series for real-time object detection, achieving unprecedented performance. This version eliminates the need for non-maximum suppression (NMS) (Rothe et al. 2015), a traditional bottleneck in earlier models, thereby drastically reducing latency. YOLOv10 introduces a dual assignment strategy in its training protocol, which optimizes detection accuracy without sacrificing speed with the help of one-to-many and one-to-one label assignments, ensuring robust detection with lower latency (Li et al. 2023; Tian et al. 2024). The architecture of YOLOv10 includes several innovative components that enhance both computational efficiency and detection performance. Among these are lightweight classification heads (Bhagat et al. 2021) that reduce computational demands, spatial-channel decoupled downsampling to minimize information loss during feature reduction (Hu et al. 2023), and rank-guided block design that optimizes parameter use (Yang et al. 2023). These architectural advancements ensure that YOLOv10 operates synergistically across various scales-from YOLOv10-N (Nano) to YOLOv10-X (Extra Large), making it adaptable to diverse computational constraints and operational requirements (Wang et al. 2024). According to wang et al. (Wang et al. 2024), performance evaluations on benchmark datasets like MS-COCO (Lin et al. 2014) demonstrate that YOLOv10 not only surpasses its predecessors-YOLOv9 and YOLOv8-in both accuracy and efficiency but also sets new industry standards. For instance, YOLOv10-S substantially outperforms comparable models (e.g., YOLOv9 BASE, YOLOV9 Gelan, YOLOv8, YOLOv7) with an improved mAP and lower latency. This version also incorporates holistic efficiency-accuracy driven design, large-kernel convolutions, and partial self-attention modules, collectively improving the trade-off between computational cost and detection capability. The architecture diagrams of YOLOv10, YOLOv9, and YOLOv8 are summarized in Figs. 8, 9, and 10, respectively.

The YOLOv10 model offers various configurations, each tailored to specific performance needs within real-time object detection frameworks. Starting with YOLOv10-N (Nano), it demonstrates a rapid detection capability with a mAP of 38.5% at an exceptionally reduced latency to 1.84 ms, making it highly suitable for scenarios demanding quick responses. Progressing through the series, YOLOv10-S (Small) and YOLOv10-M (Medium) offer progressively higher mAP values of 46.3% and 51.1% at latencies of 2.49 ms and 4.74 ms, respectively, providing a balanced performance for versatile applications. The larger variants, YOLOv10-B (Balanced) and YOLOv10-L (Large), cater to environments requiring

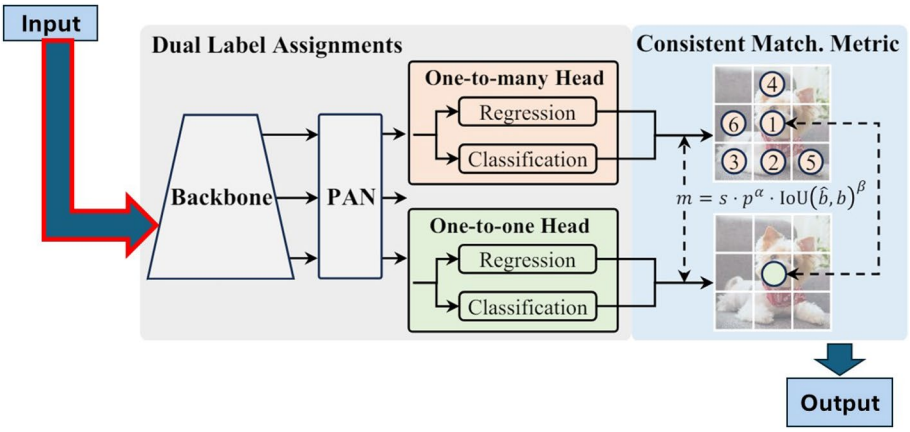


Fig. 8 YOLOv10 architecture, which employs a dual label assignment strategy to improve detection accuracy. A backbone processes the input image, while PAN (Path Aggregation Network) enhances feature representation. Employed heads are (1) one-to-many head for regression and classification tasks, and (2) one-to-one head for precise localization (Wang et al. 2024)

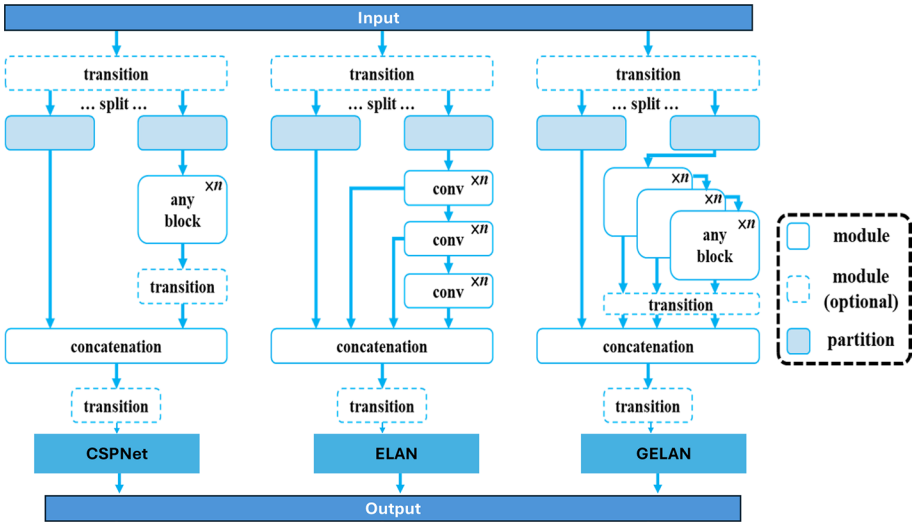


Fig. 9 YOLOv9 architecture (Wang et al. 2024) with CSPNet, ELAN, and GELAN modules. CSPNet enhances gradient flow and reduces computational load through feature map partitioning. ELAN focuses on the linear aggregation of features for improved learning efficiency, while GELAN generalizes this approach to combine features from multiple depths and pathways, providing greater flexibility and accuracy in feature extraction

detailed detections, with mAPs of 52.5% and 53.2% and latencies of 5.74 ms and 7.28 ms respectively. The largest model, YOLOv10-X (Extra Large), excels with the highest mAP of 54.4% at a latency of 10.70 ms, designed for complex detection tasks where precision is paramount. These configurations underscore YOLOv10's adaptability across a spectrum of operational requirements.

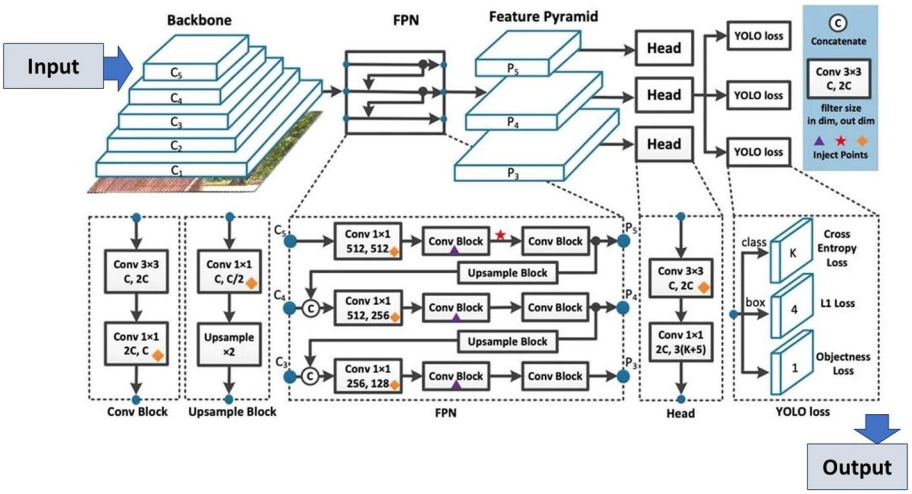


Fig. 10 YOLOv8 architecture (Jocher et al. 2022): showcasing the key components and their connections. The backbone network processes the input image through multiple convolutional layers (C1 to C5), extracting hierarchical features. These features are then passed through the Feature Pyramid Network (FPN) to create a feature pyramid (P3, P4, P5), which enhances detection at different scales. The network heads perform final predictions, incorporating convolutional blocks and upsample blocks to refine features

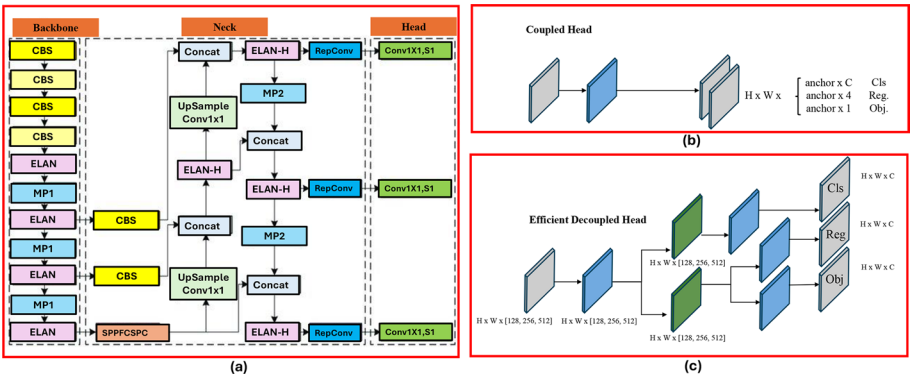


Fig. 11 Architecture diagrams for **a** YOLOv7; **b** YOLOv6; and **c** YOLOv5

Reflecting on YOLO's evolution, starting from YOLOv1, which set the benchmark with an mAP of 63.4% and a latency of 45 ms, to the latest YOLOv10, significant technological strides have been evident. YOLOv10's predecessors, YOLOv9 and YOLOv8, display comparable mAP scores to YOLOv10 but with marginally higher latency, indicating the incremental enhancements YOLOv10 brings to the table. Specifically, YOLOv9 and YOLOv8 models, such as YOLOv9-N and YOLOv8-N, showcase mAPs of 39.5% and 37.3%, respectively, at latency indicative of their generational improvements. Meanwhile, the higher end of these series, YOLOv9-X, and YOLOv8-X, achieve mAPs of 54.4% and 53.9%, respectively, with YOLOv10 outperforming them in efficiency. The YOLO series, from YOLOv1 through YOLOv8, YOLOv9, and now YOLOv10, has continually advanced the frontier of

real-time object detection, enhancing both the speed and accuracy of detections, and thus broadening the scope for practical applications in sectors like autonomous driving, surveillance, and real-time video analytics.

YOLOv9 (Wang et al. 2024) marks a significant advancement in real-time object detection by addressing the efficiency and accuracy challenges associated with earlier versions, particularly by mitigating information loss in deep neural processing. It introduces the innovative Programmable Gradient Information (PGI) and the Generalized Efficient Layer Aggregation Network (GELAN) architecture. These enhancements focus on preserving crucial information across the network, ensuring robust and reliable gradients that prevent data degradation, which is common in deep neural networks (Tishby and Zaslavsky 2015). Compared to its successor, YOLOv10, YOLOv9 sets a foundational stage by addressing the information bottleneck problem that typically hinders deep learning models. While YOLOv9's PGI strategically maintains data integrity throughout the processing layers, YOLOv10 builds upon this foundation by eliminating the need for NMS and further optimizing model architecture for reduced latency and enhanced computational efficiency. YOLOv10 also introduces dual assignment strategies for NMS-free training, significantly enhancing the system's response time without compromising accuracy, which reflects a direct evolution from the groundwork laid by YOLOv9's innovations (Zhang et al. 2023). Furthermore, YOLOv9's GELAN architecture represents a pivotal improvement in network design, offering a flexible and efficient structure that effectively integrates multi-scale features. While GELAN contributes significantly to YOLOv9's performance, YOLOv10 extends these architectural improvements to achieve even greater efficiency and adaptability (Chien et al. 2024). It reduces computational overhead and increases the model's applicability to various real-time scenarios, showcasing an advanced level of refinement that leverages and enhances the capabilities introduced by YOLOv9.

YOLOv8 was released in January 2023 by Ultralytics, marking a significant progression in the YOLO series with an introduction of multiple scaled versions designed to cater to a wide range of applications (Ultralytics 2024a, b). These versions included YOLOv8n (nano), YOLOv8s (small), YOLOv8m (medium), YOLOv8l (large), and YOLOv8x (extra-large), each optimized for specific performance and computational needs. This flexibility made YOLOv8 highly versatile, supporting many vision tasks such as object detection, segmentation, pose estimation, tracking, and classification, significantly broadening its application scope in real-world scenarios (Ultralytics 2024b). The architecture of YOLOv8 underwent substantial refinements to enhance its detection capabilities. It retained a similar backbone to YOLOv5 but introduced modifications in the CSP Layer, now evolved into the C2f module-a cross-stage partial bottleneck with dual convolutions that effectively combine high-level features with contextual information to bolster detection accuracy. YOLOv8 transitioned to an anchor-free model with a decoupled head, allowing independent processing of object detection, classification, and regression tasks, which, in turn, improved overall model accuracy (Ultralytics 2024). The output layer employed a sigmoid activation function for objectness scores and softmax for class probabilities, enhancing the precision of bounding box predictions. YOLOv8 also integrated advanced loss functions like CIoU (Du et al. 2021) and Distribution Focal Loss (DFL) (Xu et al. 2022) for bounding-box optimization and binary cross-entropy for classification, which proved particularly effective in enhancing detection performance for smaller objects. YOLOv8's architecture, demonstrated in detailed diagrams, features the modified CSPDarknet53 backbone with the innovative C2f

module, augmented by a spatial pyramid pooling fast (SPPF) layer that accelerates computation by pooling features into a fixed-size map. This model also introduced a semantic segmentation variant, YOLOv8-Seg, which utilized the backbone and C2f module, followed by two segmentation heads designed to predict semantic segmentation masks efficiently. This segmentation model achieved state-of-the-art results on various benchmarks while maintaining high speed and accuracy, evident in its performance on the MS COCO dataset where YOLOv8x reached an AP of 53.9% at 640 pixels image size-surpassing the 50.7% AP of YOLOv5-with a remarkable speed of 280 FPS on an NVIDIA A100 using TensorRT. As we progress backwards through the YOLO series, from YOLOv10 to YOLOv8 and soon to YOLOv7, these architectural and functional advancements highlight the series' evolutionary trajectory in optimizing real-time object detection networks.

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4.3 YOLOv7, YOLOv6 and YOLOv5

The YOLOv7 model introduces enhancements in object detection tailored for drone-captured scenarios, particularly through the Transformer Prediction Head (TPH-YOLOv5) variant (Zhu et al. 2021), which emphasizes improvements in handling scale variations and densely packed objects (Wang et al. 2023). By incorporating TPH and the Convolu-

tional Block Attention Module (CBAM) (Woo et al. 2018), YOLOv7 substantially boosts its capacity to focus on relevant regions in cluttered environments. These features particularly enhance the model's ability to detect objects across varied scales, an essential trait for drone applications where altitude changes affect object size perception drastically. The model integrates sophisticated strategies like multi-scale testing (Hnewa and Radha 2023) and a self-trained classifier, which refines its performance on challenging categories by specifically addressing common issues in drone imagery, such as motion blur and occlusion. These adaptations have shown notable improvements, with YOLOv7 achieving competitive results in drone-specific datasets and challenges (Bai et al. 2024). The model's adaptability and robustness in such specialized conditions demonstrate its potential beyond conventional settings, catering effectively to next-generation applications like urban surveillance and wildlife monitoring.

YOLOv6 emerges as a robust solution in industrial applications by delivering a finely balanced trade-off between speed and accuracy, crucial for deployment across various hardware platforms (Li et al. 2022). It iterates on previous versions by incorporating cutting-edge network designs, training strategies, and quantization techniques to enhance its efficiency and performance significantly. This model has been optimized for diverse operational requirements with its scalable architecture, ranging from YOLOv6-N to YOLOv6-X, each offering different performance levels to suit specific computational budgets (Sirisha et al. 2023). Significant innovations in YOLOv6 include advanced label assignment techniques and loss functions that refine the model's predictive accuracy and operational efficiency. By leveraging state-of-the-art advancements in machine learning, YOLOv6 not only excels in traditional object detection metrics but also sets new standards in throughput and latency, making it exceptionally suitable for real-time applications in industrial and commercial domains.

YOLOv6 and YOLOv7 each introduced innovative features that build on the foundation set by YOLOv5. YOLOv6, released in October 2021, introduced lightweight nano models optimized for mobile and CPU environments alongside a more effective backbone for improved small object detection. YOLOv7 further advanced this development by incorporating a new backbone network, PANet (Wang et al. 2019), enhancing feature aggregation and representation, and introducing the CIOU loss function for better object scaling and aspect ratio handling. YOLO-v6 significantly shifts the architecture to an anchor-free design, incorporating a self-attention mechanism to better capture long-range dependencies and employing adaptive training techniques to optimize performance during training (Zhang et al. 2021). These versions collectively push the boundaries of object detection performance, emphasizing speed, accuracy, and adaptability across various deployment scenarios.

YOLOv5 has significantly contributed to the YOLO series evolution, focusing on user-friendliness and performance enhancements (Ultralytics 2024a, b). Its introduction by Ultralytics brought a streamlined, accessible framework that lowered the barriers to implementing high-speed object detection across various platforms. YOLOv5's architecture incorporates a series of optimizations including improved backbone, neck, and head designs which collectively enhance its detection capabilities. The model supports multiple size variants, facilitating a broad range of applications from mobile devices to cloud-based systems (Ultralytics 2024a). YOLOv5's adaptability is further evidenced by its continuous updates and community-driven enhancements, which ensure it remains at the forefront of object detection technologies. This version stands out for its balance of speed, accuracy, and

utility, making it a preferred choice for developers and researchers looking to deploy state-of-the-art detection systems efficiently.

YOLOv5 marks a significant evolution in the YOLO series, focusing on production-ready deployments with streamlined architecture for real-world applications. This version emphasizes reducing the model's complexity by refining its layers and components, enhancing its inference speed without sacrificing detection accuracy. The backbone and feature extraction layers were optimized to accelerate processing, and the network's architecture was simplified to facilitate faster data throughput. Importantly, YOLO v5 enhances its deployment flexibility, catering to edge devices with limited computational resources through model modularity and efficient activations. These architectural refinements ensure YOLO v5 operates effectively in diverse environments, from high-resource servers to mobile devices, making it a versatile tool in the arsenal of object detection technologies.

4.4 YOLOv4, YOLOv3, YOLOv2 and YOLOv1

The introduction of YOLOv4 (Bochkovskiy et al. 2020) in 2020 marked the latest developments, employing CSPDarknet-53 (Mahasin and Dewi 2022) as its backbone. This modified version of Darknet-53 uses Cross-Stage Partial connections to reduce computational demands while enhancing learning capacity.

YOLOv4 incorporates innovative features such as Mish activation (Misra 2019), replacing traditional ReLU to maintain smooth gradients, and utilizes new data augmentation techniques such as Mosaic and CutMix (Yun et al. 2019). Additionally, it introduces advanced regularization methods, including DropBlock regularization (Ghiasi et al. 2018) and Class Label Smoothing to prevent overfitting (Müller et al. 2019), alongside optimization strategies termed BoF (Bag of Freebies) (Zhang et al. 2019) and BoS (Bag of Specials) that enhance training and inference efficiency. "YOLOv3, introduced in 2018 before the release of YOLOv4, employed the Darknet-53 architecture, incorporating principles of residual learning. Initially trained on ImageNet, this version excelled in detecting objects of various sizes due to its multi-scale detection capabilities within the architecture. The subsequent development of YOLOv4 built upon the success of YOLOv3, further enhancing the framework's robustness and accuracy.

YOLOv3 (Redmon and Farhadi 2018) improved detection accuracy, especially for small objects, by using three different scales for detection, thereby capturing essential features at various resolutions. Earlier, YOLOv2 and the original YOLO (YOLOv1) laid the groundwork for these advancements (Redmon et al. 2016).

Earlier, YOLOv2 and the original YOLO (YOLOv1) laid the groundwork for these advancements. Released in 2016, YOLOv2 introduced a new 30-layer architecture with anchor boxes from Faster R-CNN and batch normalization to speed up convergence and enhance model performance. YOLOv1, debuting in 2015 by Joseph Redmon, revolutionized object detection with its single-shot mechanism that predicted bounding boxes and class probabilities in one network pass, utilizing a simpler Darknet-19 architecture. This initial approach significantly accelerated the detection process, establishing the foundational techniques that would be refined in later versions of the YOLO series. YOLOv4 and YOLOv3, showcasing their advanced architectures and features, are illustrated in Fig. 12a and b, respectively, while YOLOv2 and YOLOv1 are depicted in Fig. 13a and b, showcasing the foundational developments in the series.

4.5 Alternative versions derived from YOLO

Several alternative YOLO models have been developed from different versions, with the five primary ones being YOLO-NAS, YOLO-X, YOLO-R, DAMO-YOLO, and Gold-YOLO.

4.5.1 YOLO-NAS

YOLO-NAS, developed by Deci AI, represents a significant advancement in object detection technology (Terven et al. 2023). This model leverages Neural Architecture Search (NAS) (Ren et al. 2021) to address limitations of previous YOLO iterations such as YOLOv4, YOLOv5, YOLOv6 and YOLOv7 (Mithun and Jawhar 2024). YOLO-NAS introduces a quantization-friendly basic block, enhancing performance with minimal precision loss post-quantization. The architecture employs quantization-aware blocks and selective quantization, resulting in superior object detection capabilities. Notably, when converted to INT8, the model experiences only a slight precision drop, outperforming its predecessors. YOLO-NAS utilizes sophisticated training schemes and post-training quantization techniques, further improving its efficiency (Terven et al. 2023). The model is pre-trained on datasets such as COCO, Objects365, and Roboflow 100, making it suitable for various downstream object detection tasks. YOLO-NAS is available in three variants: Small (s), Medium (m), and Large (l), each optimized for different computational requirements. These variants offer a balance between Mean Average Precision (mAP) and latency, with the INT-8 versions demonstrating impressive performance metrics. The architecture of YOLO-NAS

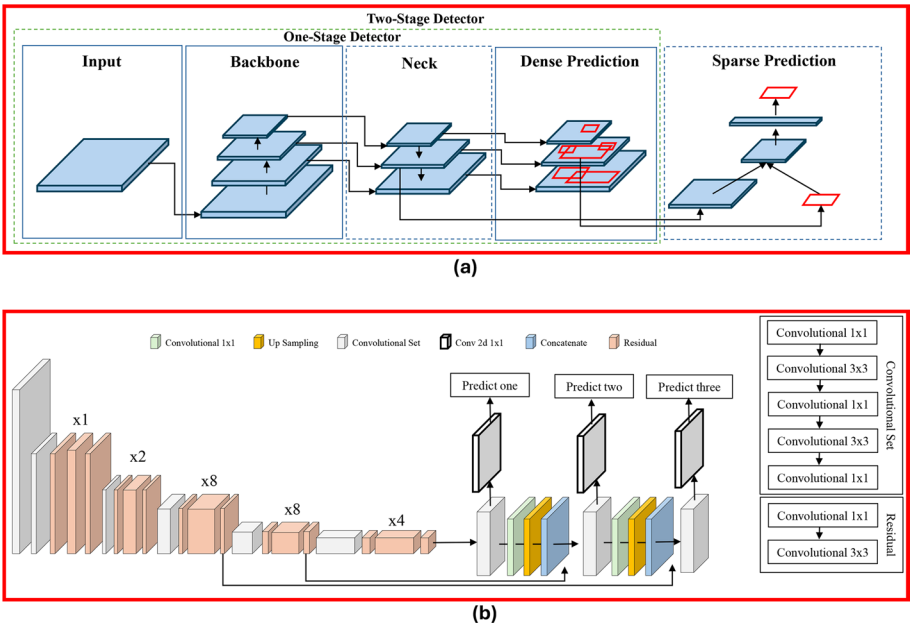


Fig. 12 Comparison of YOLOv4 (Bochkovskiy et al. 2020) and YOLOv3 (Redmon and Farhadi 2018) architectures. **a** YOLOv4 architecture shows a two-stage detector with a backbone, neck, dense prediction, and sparse prediction modules. **b** YOLOv3 architecture features convolutional and upsampling layers that lead to multi-scale predictions. This highlights the structural advancements in object detection between the two versions

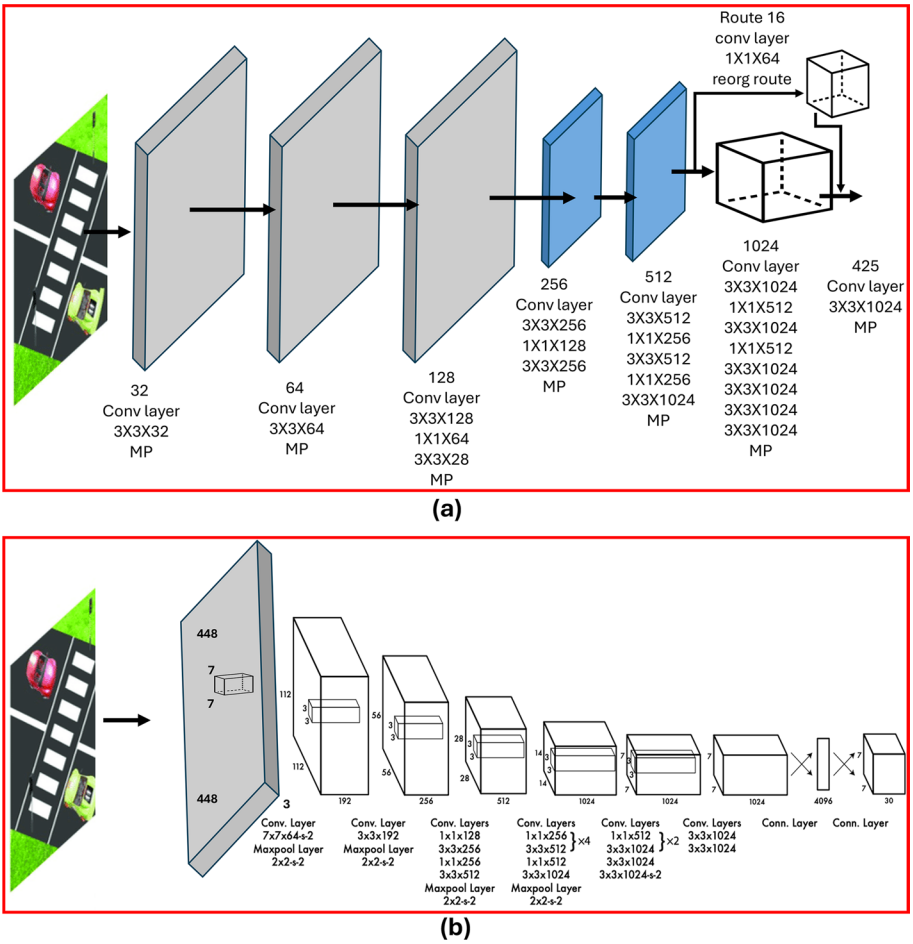


Fig. 13 **a** YOLOv2 architecture (Redmon and Farhadi 2017), illustrating improvements such as the use of batch normalization, higher resolution input, and anchor boxes; **b** YOLOv1 architecture (Redmon et al. 2016), showing the sequence of convolutional layers, max-pooling layers, and fully connected layers used for object detection. This model performs feature extraction and prediction in a single unified step, aiming for real-time performance

(Fig. 14a) supports inference, validation, and export modes, though it does not support training. YOLO-NAS's innovative design and superior performance position it as a critical tool for developers and researchers in the field of computer vision.

4.5.2 YOLO-X

YOLOX, developed by Megvii Technology, represents a significant advancement in the YOLO series of object detectors. This model introduces several key improvements to enhance performance and efficiency. YOLOX adopts an anchor-free approach, departing from the anchor-based methods of its predecessors such as YOLOv6, YOLOv7 and YOLOv8 (Ge 2021). It incorporates a decoupled head, separating classification and regres-

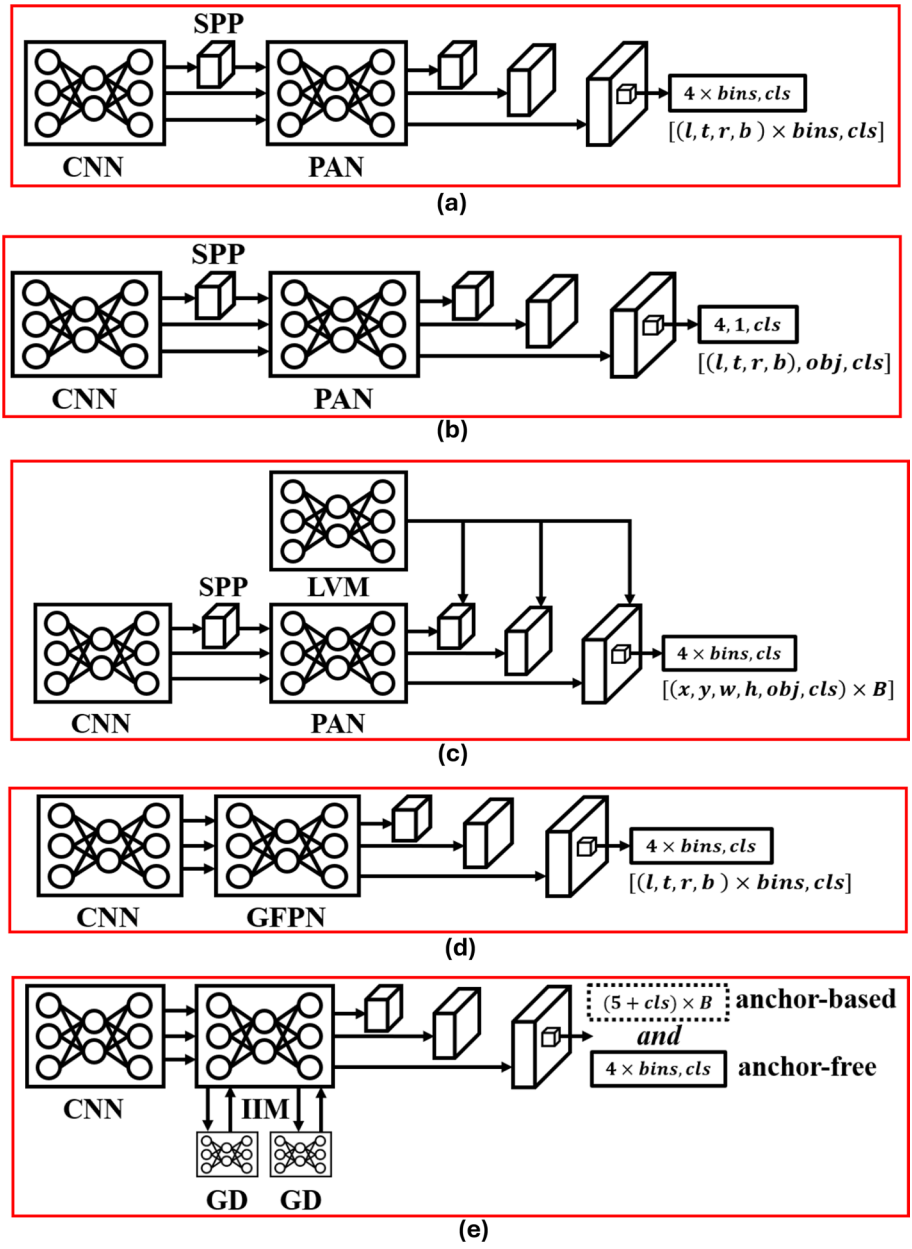


Fig. 14 Architecture diagram of **a** YOLONAS; **b** YOLOX; **c** YOLOR; **d** DAMO YOLO; **e** GOLD YOLO

sion tasks to address the known conflict between these objectives in object detection (Zhang et al. 2022). The model also implements SimOTA, an advanced label assignment strategy, further improving its detection capabilities (Liu and Sun 2022). Architecturally (Fig. 14b), YOLOX-DarkNet53 builds upon the YOLOv3-SPP baseline, incorporating enhancements such as EMA weights updating, cosine learning rate scheduling, IoU loss, and an IoU-aware

branch. The decoupled head consists of a 1x1 convolution layer for channel dimension reduction, followed by two parallel branches with two 3x3 convolution layers each (Ashraf et al. 2024). This design significantly improves convergence speed and is crucial for end-to-end detection performance. YOLOX demonstrates superior performance across various model sizes. The YOLOX-L variant achieves 50.0% AP on COCO at 68.9 FPS on Tesla V100, surpassing YOLOv5-L by 1.8% AP. Even the lightweight YOLOX-Nano, with only 0.91M parameters and 1.08 GFLOPs, attains 25.3% AP on COCO, outperforming NanoDet by 1.8% AP. These advancements position YOLOX as a state-of-the-art object detector, balancing accuracy and efficiency across a wide range of model scales (Ge 2021).

4.5.3 YOLO-R

YOLOR (You Only Learn One Representation) is a novel object detection algorithm developed by Chang et al. (2023). Unlike other YOLO versions, YOLOR introduces a novel approach to multi-task learning by unifying implicit and explicit knowledge representation (Andrei-Alexandru et al. 2022). The algorithm's core concept is inspired by human cognition, aiming to process multiple tasks simultaneously given a single input. YOLOR's architecture (Fig. 14c) incorporates three key components: kernel space alignment, prediction refinement, and a CNN with multi-task learning capabilities. This unified network encodes both implicit knowledge (learned subconsciously from deep layers) and explicit knowledge (obtained from shallow layers and clear metadata), resulting in a more refined and generalized representation (Sun et al. 2024; Chang et al. 2023). Compared to other YOLO algorithms such as YOLOv9, YOLOv5 or YOLOv3, YOLOR significantly improves both speed and accuracy. It achieves comparable object detection accuracy to Scaled YOLOv4 while increasing inference speed by 88%, making it one of the fastest object detection algorithms in modern computer vision. On the MS COCO dataset, YOLOR outperforms PP-YOLOv2 by 3.8% in mean average precision at the same inference speed (Chang et al. 2023).

4.5.4 DAMO-YOLO

DAMO-YOLO is developed by Alibaba's DAMO Academy, which significantly enhances performance by integrating novel technologies like Neural Architecture Search (NAS), a reparameterized Generalized-FPN (RepGFPN), and lightweight head architectures (Fig. 14d) with AlignedOTA label assignment and distillation enhancement (Xu et al. 2022). Leveraging MAE-NAS, the model employs a heuristic, training-free approach to architect detection backbones under strict latency and performance constraints, generating efficient structures akin to ResNet and CSPNet (Terven et al. 2023). The neck design emphasizes a robust "large neck, small head" architecture, optimizing the fusion of high-level semantic and low-level spatial features through an enhanced FPN. This approach effectively balances computational efficiency and detection accuracy, particularly notable in its deployment across various model scales, from lightweight versions for edge devices to more robust configurations for general industry applications. DAMO-YOLO's architectural prowess is showcased through impressive performance metrics, achieving mAP scores ranging from 43.6 to 51.9 on COCO datasets with relatively low latency on T4 GPUs. Moreover, the model's lightweight variants demonstrate substantial efficiency on edge devices, underscoring its adaptability and broad application potential. Such capabilities are further augmented by

strategic enhancements in label assignment and knowledge distillation, addressing common challenges in object detection like label misalignment and model generalization.

4.5.5 Gold-YOLO

Gold-YOLO was developed by the team at Huawei Noah's Ark Lab to significantly enhance multi-scale feature fusion through an innovative Gather-and-Distribute (GD) mechanism (Wang et al. 2024). This mechanism, which utilizes convolution and self-attention operations, was implemented to optimize the exchange and integration of information across different levels of the feature pyramid. This approach facilitated a more effective balance between latency and detection accuracy (Wang et al. 2024). Furthermore, an MAE-style unsupervised pretraining was incorporated into the YOLO-series for the first time, which was reported to enhance learning efficiency and overall model performance. Gold-YOLO's architecture (Fig. 14e) aimed to address the limitations inherent in traditional Feature Pyramid Networks (FPNs) by preventing recursive information loss and enabling more direct and efficient feature fusion. This was achieved by a method where features from all levels were gathered to a central processing node, enhanced, and then redistributed, ensuring enriched feature maps that retained critical information across scales. The effectiveness of this novel design was demonstrated through good performance metrics; Gold-YOLO achieved a 39.9% AP on the COCO dataset with high throughput speeds on a T4 GPU, surpassing previous state-of-the-art models like YOLOv6−3.0-N. The contributions made by this paper were significant, as they not only enhanced the YOLO model's capabilities to handle various object sizes and complexities but also established a new benchmark for the integration of advanced neural network techniques with traditional convolutional architectures for real-time applications.

5 Applications

YOLO has many real-time practical applications (Vijayakumar and Vairavasundaram 2024; Chen et al. 2023), such as autonomous vehicles and traffic safety where the technology is used for obstacle detection, pedestrian pose estimation for intention prediction, and traffic sign recognition, enhancing safety and navigation (Gheorghe et al. 2024). Similarly, YOLO is employed in healthcare for detecting anomalies in medical images, aiding in accurate and efficient diagnostics (Vijayakumar and Vairavasundaram 2024; Ragab et al. 2024). Additionally, industrial manufacturing benefits from YOLO's capabilities, with applications in quality control and defect detection (Li et al. 2022; Hussain 2023), in surveillance for intrusion detection and anomaly identification (Mohod et al. 2022), while in agriculture, it supports crop stress detection, monitoring, and robotic fruit harvesting, among other use cases (Alibabaei et al. 2022; Badgujar et al. 2024; Wang et al. 2021).

The remainder of this section is categorically divided into five key application areas where YOLO models have demonstrated significant impact: Autonomous Vehicles and Traffic Safety, Healthcare and Medical Imaging, Security and Surveillance, Industrial Manufacturing, and Agriculture.

5.1 Autonomous vehicles and traffic safety

Each YOLO version has been pivotal in advancing the capabilities of autonomous vehicles and traffic safety by providing highly efficient and accurate real-time detection systems. Each iteration of YOLO has brought improvements that enhance the vehicle's ability to perceive its environment quickly and accurately, which is critical for safe navigation and decision-making (Benjumea et al. 2021; Malligere Shivanna and Guo 2024). Starting with YOLOv1 (Redmon et al. 2016), the YOLO algorithm revolutionized the approach by performing detection tasks directly from full images in a single network pass, allowing for the detection of objects at a remarkable speed (Sarda et al. 2021). This initial model was pivotal, setting a high standard for real-time object detection and establishing a framework that future versions would build upon. Subsequent iterations, including YOLOv2 and YOLOv3, continued to refine this approach by introducing concepts such as real-time multi-scale processing and improved anchor box adjustments, which enhanced the accuracy and robustness of the detections. These versions were particularly adept at handling the variable scales of objects seen in driving environments-from nearby pedestrians to distant road signs-making them invaluable for autonomous driving applications. YOLOv4 and later versions further pushed the boundaries by integrating advanced neural network techniques and optimizations that improved detection accuracy while maintaining the high-speed processing necessary for real-time applications (Cai et al. 2021; Zhao et al. 2022). These advancements in YOLO technology have not only bolstered the capabilities of autonomous vehicles in terms of environmental perception and decision-making but have also significantly contributed to advancements in automotive safety and operational reliability (Woo et al. 2022).

Ye et al. (2022) developed an end-to-end adaptive neural network control for autonomous vehicles that predicts steering angles using YOLOv5, enhancing vehicle navigation precision (Ye et al. 2022). Mostafa et al. (2022) compared the effectiveness of YOLOv5, YOLOX, and Faster R-CNN in detecting occluded objects for autonomous vehicles, improving detection reliability (Mostafa et al. 2022). Jia et al. (2023) proposed an enhanced YOLOv5 detector for autonomous driving, which offers increased speed and accuracy (Jia et al. 2023). Chen et al. (2023) utilized an improved YOLOv5-OBB algorithm for autonomous parking space detection in electric vehicles, enhancing operational efficiency (Chen et al. 2023). Liu and Yan (2022) customized YOLOv7 for vehicle-related distance estimation, providing essential metrics for safe navigation (Liu and Yan 2022). Mehla et al. (2023) evaluated YOLOv8 against EfficientDet in autonomous maritime vehicles, highlighting the superior detection capabilities of YOLOv8 (Mehla et al. 2023).

Further advancements with YOLOv8 have led to significant improvements in object detection in adverse weather conditions, an area of particular concern for autonomous driving. Applying transfer learning techniques using datasets from diverse weather conditions has markedly increased the detection performance of YOLOv8, ensuring reliable recognition of crucial road elements like pedestrians and obstacles under challenging weather scenarios (Kumar and Muhammad 2023). Additionally, the development of YOLOv8 for specific tasks such as brake light status detection illustrates the algorithm's flexibility and its potential in enhancing interpretability and safety for autonomous vehicles (Oh and Lim 2023). These innovations underscore the critical role of YOLOv8 and YOLOv9 in pushing the boundaries of what is possible in the autonomous vehicle industry, highlighting their impact in meeting the rigorous demands for safety and reliability in self-driving technolo-

gies (Afdhal et al. 2023). YOLOv8 and YOLOv9 are at the forefront of transforming the landscape of autonomous vehicle technologies, playing a pivotal role in enhancing the operational safety and efficiency of self-driving cars. These models have excelled in real-time object detection, a crucial aspect of autonomous driving, especially under the challenging and variable conditions typical in real-world traffic environments. An enhanced version, addresses the need for detecting smaller objects such as traffic signs and signals, demonstrating its utility with a notable accuracy rate and efficiency in processing, making it ideal for high-speed driving scenarios (Wang et al. 2024). Table 2 illustrates different applications of YOLO in the autonomous vehicle industry, presented in reverse chronological order from the most recent versions to the older ones.

5.1.1 Pedestrian pose estimation

Ali et al. (2023) presented a Bayesian Generalized Extreme Value Model to evaluate real-time pedestrian crash risks at signalized intersections, leveraging advanced AI-based video analytics. This framework employs deep learning algorithms like YOLO for precise object detection and DeepSORT for effective tracking. The model concentrates on crucial safety indicators such as Post Encroachment Time (PET). Through this approach, the study underscores the significant role of AI-driven video analysis in boosting intersection safety by delivering real-time risk assessments. This development signifies a substantial advancement in the proactive management of traffic safety. Hussain et al. (2024) explored the enhancement of pedestrian crash estimation using machine learning techniques focused on anomaly detection. Their study addresses the limitations of traditional Extreme Value Theory (EVT) models by applying unconventional sampling methods, thereby increasing the accuracy and reducing uncertainty in crash risk estimations. The use of YOLO for object detection and DeepSORT for tracking is pivotal in this methodology, enhancing detection accuracy and tracking reliability in real-time scenarios. Ghaziamin et al. (2024) developed a privacy-preserving real-time passenger counting system for bus stops using overhead fisheye cameras. This innovative system employs YOLOv4, Detecnet-V2 and Faster-RCNN for detection purposes and DeepSORT for tracking. The system processes data in real-time at 30 frames per second (FPS) when utilizing YOLOv4 as the detection model. This technology significantly enhances transit planning by providing accurate passenger counts, while also maintaining passenger privacy and energy efficiency.

Additionally, Pedestrian-vehicle conflict prediction was explored by Zhang et al. (2020) proposed a model employing a Long Short-Term Memory (LSTM) neural network to forecast pedestrian-vehicle conflicts at signalized intersections by analyzing video data. The model uses YOLOv3 for object detection and Deep SORT for tracking, achieving impressive accuracy rates and demonstrating the transformative potential of LSTM networks in collision warning systems. This approach suggests a proactive enhancement of pedestrian safety in connected vehicle environments.

Crossing intention prediction and behavioral analysis was explored by Zhang et al. (2020) utilized an LSTM neural network to predict pedestrian red-light crossing intentions at intersections by analyzing video data of real traffic scenarios. The model uses YOLOv3 for detection and DeepSORT for tracking, recognizing patterns that indicate potential red-light crossings with a high accuracy rate. This capability aims to improve traffic safety through vehicle-to-infrastructure communication systems that alert drivers to potential pedestrian

Table 2 Studies on YOLO applications focus on object detection and real-time performance improvements to enhance autonomous vehicles and traffic safety

Title of paper	Description of work	Purpose and YOLO usage	Version	Refs. and Year
"Multi-Class Vehicle Detection and Classification with YOLO11 on UAV-Captured Aerial Imagery"	Utilizes YOLO11 for real-time vehicle detection and classification in UAV-captured traffic images, with a comparative analysis against YOLOv10	Focuses on robust vehicle detection in dynamic UAV-captured images, enhancing real-time processing and accuracy in aerial traffic monitoring	YOLO11	Bakirci et al. (2024), 2024
"YOLOv10-Based Real-Time Pedestrian Detection for Autonomous Vehicles"	Presents a real-time pedestrian detection method enhancing YOLOv10 with EfficientNet backbone, C2F-DM, BiFormer, and multi-scale feature fusion detection head for complex environments	Aims to enhance autonomous vehicle safety by providing efficient multi-scale pedestrian detection	YOLOv10	Li et al. (2024), 2024
"Improved YOLOv10 for Visually Impaired: Balancing Model Accuracy and Efficiency in the Case of Public Transportation"	Introduces Improved-YOLOv10 with Coordinate Attention and Adaptive Kernel Convolution for enhanced bus detection and POV classification for the visually impaired	Enhances accessibility in public transportation for visually impaired individuals through improved detection and efficiency	YOLOv10	Arifando et al. (2025), 2025
"Transforming Aircraft Detection Through LEO Satellite Imagery and YOLOv9 for Improved Aviation Safety"	Utilizes YOLOv9 with LEO satellite imagery for enhanced aircraft detection in wide-area airport environments	Aims to improve airport security and aviation safety by integrating advanced YOLO-based object detection with satellite imagery	YOLOv9	Bakirci and Bayraktar (2024), 2024
"YOLOv8-QSD: An Improved Small Object Detection Algorithm for Autonomous Vehicles Based on YOLOv8"	Developed an anchor-free, BiFPN-enhanced YOLOv8 model for better small object detection in driving scenarios	Enhances detection of small objects for autonomous vehicles with reduced computational demands, tested on SODA-A dataset	YOLOv8-QSD	Wang et al. (2024), 2024
"Object Detection in Dense and Mixed Traffic for Autonomous Vehicles With Modified YOLO"	Adapted YOLOv7 with deformable layers and softNMS for object detection in heavy Indonesian traffic	Enhances detection and classification of objects around autonomous vehicles using a modified YOLOv7, tested on a novel Indonesian traffic dataset	YOLOv7-MOD	Wibowo et al. (2023), 2023
"Object Tracking for Autonomous Vehicle Using YOLOV3"	Evaluated YOLOv3 for object tracking in autonomous vehicles	Two models were provided, one trained using only the online COCO dataset and the other with additional images from various locations at Universiti Malaysia Pahang (UMP).	YOLOv3	Hung et al. (2022), 2022
"The improvement in obstacle detection in autonomous vehicles using YOLO non-maximum suppression fuzzy algorithm"	Employed a hybrid of fuzzy logic and NMS in YOLO for better obstacle detection in autonomous driving	Enhances obstacle detection accuracy and speed using a modified YOLO algorithm	YOLOv3	Zaghari et al. (2021), 2021

violations, thereby preventing accidents. Yang et al. (2022) introduced the VENUS smart node, a cooperative traffic signal assistance system for non-motorized users and individuals with disabilities. This novel infrastructure leverages computer vision and edge AI to integrate real-time data on pedestrian movement and intent. The system employs YOLOv4 for detection and OpenPose for pose estimation, achieving high accuracy in detecting crossing intentions and mobility status across various test sites. This innovation has significant potential for widespread use in smart city infrastructures, greatly enhancing safety and accessibility.

Jiao and Fei (2023) conducted a study on monitoring pedestrian walking speeds at the street level using drones. The research utilized UAV-based video footage to measure the walking speeds of pedestrians on a commercial street. Deep learning algorithms, particularly YOLOv5 for object detection and DeepSORT for tracking, were employed in this study. Speed calculations were adjusted for geometric distortions using the SIFT and RANSAC algorithms, achieving high accuracy. The study found that 90.5% of the corrected speeds had an absolute error of less than 0.1 m/s, providing a precise and non-intrusive method for analyzing pedestrian walking speeds. Wang et al. (2024) used drone-captured video footage to examine "safe spaces" for pedestrians and e-bicyclists at urban crosswalks. The study discovered that e-bicyclists maintain larger semi-elliptical safe zones that are sensitive to speed changes compared to the semi-circular zones maintained by pedestrians. By quantifying these safe spaces and examining variations due to speed and traffic presence, the study offers valuable insights for enhancing crosswalk safety and managing urban traffic more effectively. The use of YOLOv3 for object detection and DeepSORT for tracking plays a critical role in this analysis. Zhou et al. (2023) developed an innovative model that integrates a pedestrian-centric environment graph with Graph Convolutional Networks (GCNs) and a pedestrian-state encoder. This model effectively captures dynamic interactions between pedestrians and their environments, providing advanced safety warnings by predicting crossing intentions up to three seconds in advance. This model holds significant potential for applications in intelligent transportation systems. The integration of YOLOv5 for detection, DeepSORT for tracking, and HRNet for pose estimation enhances the model's predictive accuracy and real-time application. Table 3 illustrates different applications of YOLO usage in pedestrian pose estimation, for intention prediction and behavioral analysis.

5.1.2 Traffic sign detection

Traffic sign detection and recognition systems play a pivotal role in enhancing road safety and are essential for the advancement of autonomous driving. These systems enable drivers, or autonomous vehicles, to effectively respond to road conditions, ensuring the safety of all road users (Flores-Calero et al. 2024). However, the complexity and variability of traffic environments, such as adverse weather conditions, combined with the small size of traffic signs present significant challenges for accurately detecting small traffic signs in real-world scenarios (Li et al. 2023; Mahaur and Mishra 2023; Zhang et al. 2020)

For instance, Li et al. (2022) presented a classical YOLO-based architecture for traffic sign recognition. First, traffic signs are categorized and preprocessed according to their specific characteristics. The processed images are then input into an optimized convolutional neural network for finer category classification. The proposed recognition algorithm was tested using a dataset based on the German traffic sign recognition standard, and its per-

Table 3 Studies on YOLO usage in pedestrian pose estimation, for intention prediction and behavioral analysis

Title of paper	Description of work	Purpose and YOLO usage	Version	Refs. and Year
"Multi-Object Pedestrian Tracking Using Improved YOLOv8 and OC-SORT"	Proposes a comprehensive approach for pedestrian tracking by combining the improved YOLOv8 object detection algorithm with the OC-SORT tracking algorithm, integrating advanced techniques such as SoftNMS, GhostConv, and C3Ghost Modules	Aimed to enhance multi-object pedestrian tracking for autonomous driving systems by improving detection accuracy and model efficiency with YOLOv8, and integrating it with the OC-SORT tracking algorithm for robust tracking in challenging scenarios	YOLOv8	Xiao and Feng (2023), 2023
"Revisiting The Hybrid Approach of Anomaly Detection and Extreme Value Theory for Estimating Pedestrian Crashes Using Traffic Conflicts Obtained from AI-Based Video Analytics"	Focuses on improving pedestrian crash predictions by utilizing machine learning for anomaly detection and integrating it with Extreme Value Theory (EVT). The approach leverages YOLOv7 for extracting traffic conflicts and relevant data, providing a robust framework for predicting pedestrian crashes from traffic incidents	YOLOv7 is used for the extraction of traffic conflicts and relevant data needed for the EVT model	YOLOv7	Hussain et al. (2024), 2024
"Pedestrian Walking Speed Monitoring at Street Scale by An In-Flight Drone"	This study introduces a method for measuring pedestrian walking speeds on a commercial street using drone video. Pedestrians are detected and tracked using YOLOv5 and the DeepSORT algorithm, with distance calculations performed using Scale-Invariant Feature Transform (SIFT) and random sample consensus (RANSAC) algorithms, followed by geometric correction	Provide an accurate and cost-effective method for monitoring pedestrian walking speeds over large areas. YOLOv5 is utilized for pedestrian detection in drone footage, which is crucial for tracking and speed calculation	YOLOv5	Jiao and Fei (2023), 2023
"Pedestrian Crossing Intention Prediction From Surveillance Videos For Over-The-Horizon Safety Warning"	Prediction of pedestrian crossing intentions using surveillance camera footage. The framework constructs a pedestrian-centric environment graph, uses a Graph Convolutional Network (GCN) for environment encoding, and employs a pedestrian-state encoder for extracting behaviour features. An intention prediction decoder is then used to determine crossing probabilities	YOLOv5 is used to identify and track pedestrians within the surveillance footage, facilitating the extraction of visual and behavioural features necessary for the prediction framework	YOLOv5	Zhou et al. (2023), 2024
"A Bayesian Generalised Extreme Value Model to Estimate Real-Time Pedestrian Crash Risks at Signalised Intersections Using Artificial Intelligence-Based Video Analytics"	Develop a Bayesian Generalised Extreme Value (GEV) model for estimating real-time pedestrian crash risks at signalized intersections. The model uses AI-based video analytics to identify and track vehicles and pedestrians, extracting traffic conflicts and relevant covariates for risk assessment	Identify crash-prone conditions and implement timely risk mitigation strategies to enhance pedestrian safety at intersections. YOLOv4 is used to detect and track pedestrians and vehicles, facilitating the extraction of necessary data for the Bayesian EVT model	YOLOv4	Ali et al. (2023), 2022

Table 3 (continued)

Title of paper	Description of work	Purpose and YOLO usage	Version	Refs. and Year
"A Privacy-Preserving Edge Computing Solution For Real-Time Passenger Counting At Bus Stops Using Overhead Fisheye Camera"	Development of a privacy-preserving edge computing solution for real-time passenger counting at bus stops using fisheye camera footage. The system leverages YOLOv4 for passenger detection and is designed to operate efficiently on edge devices powered by solar panels	YOLOv4 creates an automated, efficient, and privacy-respecting passenger counting system suitable for smart city bus stops. Model is identified as the superior object detection model in this study, outperforming DetectNet and Faster-RCNN	YOLOv4	Ghaziamin et al. (2024), 2024
"Cooperative Traffic Signal Assistance System for Non-Motorized Users and Disabilities Empowered by Computer Vision and Edge Artificial Intelligence"	Create the VENUS smart node, a system designed to aid non-motorized and disabled users at intersections. The system integrates computer vision and edge AI technologies for object recognition, user localization, and pose direction estimation, ensuring accurate detection and tracking of various users	Enhancing traffic signal assistance for non-motorized and disabled users by providing real-time data and interaction capabilities. YOLOv4 is used for object recognition, user localization, and pose direction estimation of non-motorized users	YOLOv4	Yang et al. (2022), 2022
"Examining Safe Spaces for Pedestrians and E-Bicyclists at Urban Crosswalks: An Analysis Based on Drone-Captured Video"	Assess the safety zones for pedestrians and e-bicyclists at urban crosswalks using drone footage. YOLOv3 is employed for accurate identification and tracking of individuals, allowing for detailed analysis of their movements and interactions	Enhancing the safety of pedestrians and e-bicyclists at crosswalks by analyzing their safe spaces. YOLOv3 is used for the study of user's dynamic interactions	YOLOv3	Wang et al. (2024), 2024
"Forecast Pedestrian-Vehicle Collisions at Traffic Lights"	Implements YOLOv3 for detecting pedestrian-vehicle interactions and classifying them into safe interactions, slight conflicts, and severe conflicts	Enhance pedestrian safety at intersections by modelling and predicting potential pedestrian-vehicle conflicts	YOLOv3	Zhang et al. (2020), 2020
"Prediction of Pedestrian Crossing Intentions at Intersections Based On Long Short-Term Memory Recurrent Neural Network"	Prediction pedestrian red-light crossing behaviour at intersections using LSTM networks. YOLOv3 is employed to detect pedestrians and extract relevant characteristics from video data, which are then used for behavioural prediction	YOLOv3 is used to identify pedestrians and extract relevant characteristics from the video data, which are then passed into the LSTM neural network for prediction	YOLOv3	Zhang et al. (2020), 2020

formance was compared with other baseline algorithms. Results show that the algorithm significantly improves processing speed while maintaining high classification accuracy, making it better suited for traffic sign recognition systems.

Zhang et al. (2017) presented a Chinese traffic sign detection algorithm based on a deep convolutional network. To enable real-time detection, they proposed an end-to-end convolutional network inspired by YOLOv2. Considering the characteristics of traffic signs, they incorporated multiple 1×1 convolutional layers in the intermediate network layers while reducing the number of convolutional layers in the top layers to decrease computational complexity. For effective small traffic sign detection, the input images are divided into dense grids to capture finer feature maps. Additionally, they expanded the Chinese Traffic Sign Dataset (CTSD) and enhanced the marker information available online. Experimental results using both the expanded CTSD and the German Traffic Sign Detection Benchmark (GTSDB) demonstrate that the proposed method is faster and more robust.

On the other hand, Zhang et al. (2020) proposed a new detection scheme, MSA_YOLOv3, for accurate real-time localization and classification of small traffic signs. The approach begins with data augmentation using image mixup technology. A multi-scale spatial pyramid pooling block is incorporated into the Darknet53 network, enabling more comprehensive learning of object features. Additionally, a bottom-up augmented path is designed to enhance the feature pyramid in YOLOv3, allowing effective utilization of fine-grained features in the lower layers for precise object localization. Tests on the TT100K dataset show that MSA_YOLOv3 outperforms YOLOv3 in detecting small traffic signs.

Recently, later versions of YOLO are being applied to detect traffic signs. For example, Mahaur and Mishra (2023) presented a new version called iS-YOLOv5 model, which increases the mean Average Precision (mAP) by 3.35% on the BDD100K dataset to detect traffic sign and traffic lights. While, Bai et al. (2023) introduced two innovative traffic sign detection models, called YOLOv5-DH and YOLOv5-TDHSA, based on the YOLOv5s model with the following modifications (YOLOv5-DH uses only the second modification): (1) replacing the last layer of the 'Conv + Batch Normalization + SiLU' (CBS) structure in the YOLOv5s backbone with a transformer self-attention module (T in the YOLOv5-TDHSA's name), and also adding a similar module to the last layer of its neck, so that the image information can be used more comprehensively, (2) replacing the YOLOv5s coupled head with a decoupled head (DH in both models' names) to increase the detection accuracy and speed up the convergence, and (3) adding a small-object detection layer (S in the YOLOv5-TDHSA's name) and an adaptive anchor (A in the YOLOv5-TDHSA's name) to the YOLOv5s neck to improve the detection of small objects. Their experiments were conducted using the TT100K dataset.

Similarly, Li et al. (2023) proposed a small object detection algorithm for traffic signs based on the improved YOLOv7 called SANO-YOLOv7. First, the small target detection layer in the neck region was added to augment the detection capability for small traffic sign targets. Simultaneously, the integration of self-attention and convolutional mix modules (ACmix) was applied to the newly added small target detection layer, enabling the capture of additional feature information through the convolutional and self-attention channels within ACmix. Furthermore, the feature extraction capability of the convolution modules was enhanced by replacing the regular convolution modules in the neck layer with omni-dimensional dynamic convolution (ODConv). To further enhance the accuracy of small target detection, the normalized Gaussian Wasserstein distance (NWD) metric was introduced

to mitigate the sensitivity to minor positional deviations of small objects. SANO-YOLOv7 was trained on the highly challenging TT100K public dataset.

More interesting traffic sign detection studies were presented with YOLOv8 and later YOLO versions. In the Robotaxi-Full Scale Autonomous Vehicle Competition, YOLOv8 was specifically adapted to recognize and interpret traffic signs, providing real-time alerts that are essential for safe driving (Soylu and Soylu 2024). Next, Zhang (2024) introduced an enhanced traffic sign detection algorithm based on YOLOv9. AKConv replaces the Conv module in RepNCSPELAN4, maintaining detection accuracy while reducing weight. Focal-EIoU Loss replaces the original regression loss function, Clou Loss, accelerating convergence and improving accuracy by dividing the aspect ratio's loss term into the difference between the minimum outer frame's width and height and the predicted width and height. Additionally, the network's feature extraction capability and detection accuracy are further strengthened by incorporating the Convolutional Block Attention Module (CBAM) attention mechanism. The public TT100K traffic sign dataset was used for training and evaluation.

Up until now, YOLO versions 10, 11 and 12 have not been applied in detecting traffic signs. This presents an exciting opportunity for future research and application, as these versions may offer improvements in accuracy, speed, and adaptability for detecting traffic signs in complex environments (4).

5.2 Healthcare and medical imaging

YOLO has marked a significant technological advancement in healthcare applications as well, especially with the introduction of newer versions such as YOLOv7 and YOLOv8 (Pandey et al. 2023; Ju and Cai 2023; Inui et al. 2023). The recent iterations of YOLO, particularly YOLOv7, YOLOv8, and YOLOv9, could significantly enhance medical diagnostics by offering advanced computational efficiency and improved feature extraction capabilities, making them suitable for real-time medical imaging applications. Such capabilities are crucial in urgent care scenarios, where swift diagnosis can be pivotal. For instance, YOLOv8's sophisticated algorithms excel in accurately delineating complex biological structures, vital for identifying pathologies in conditions like vascular diseases or tumors. Similarly, YOLOv9's rapid processing power enables immediate analysis of medical images, essential in emergency medical responses where timely intervention is critical. These versions have the potential to revolutionize healthcare by facilitating early detection of diseases and supporting continuous patient monitoring, transforming the traditional approach of healthcare diagnostics into one that integrates accurate, swift diagnostics seamlessly with routine medical examinations. Unlike the traditional methods which depend heavily on manual annotation and are prone to errors and subjectivity, YOLO algorithms automate the detection and localization of medical anomalies such as tumors, lesions, and other pathological markers across various imaging modalities. This automation is driven by YOLO's unique architecture that efficiently predicts multiple bounding boxes and class probabilities in a single analysis, enhancing diagnostic accuracy and reducing the potential for human error.

In the field of medical imaging and diagnostics, the adoption of the YOLO object detection algorithm has showcased promising improvements in accuracy and efficiency, particularly with its latest versions like YOLOv5, YOLOv6, YOLOv7, and YOLOv8. For instance,

Table 4 Studies on YOLO usage in traffic sign detection and recognition

Title of paper	Description of work	Purpose and YOLO usage	Version	Refs. and Year
"FINE-GRAINED CLASSIFICATION OF MILITARY AIRCRAFT USING PRE-TRAINED DEEP LEARNING MODELS AND YOLO11"	Examines the use of pre-trained CNN models and YOLO11 for fine-grained classification of military aircraft, using a dataset of 24,164 images for training and 6,042 for testing	Aims to enhance military aviation applications by achieving accurate aircraft identification and tail number extraction using advanced AI-powered image recognition	YOLO11x-cls	Karaca and Atasoy (2025), 2025
"Research on traffic sign detection based on improved YOLOv9"	An enhanced traffic sign detection algorithm based on YOLOv9 was introduced to address issues of low accuracy, missed detection, and the omission of small targets in complex environments. The TT100K traffic sign dataset was used for training and evaluation	YOLOv9 was modified by replacing the Conv module with AKConv, reducing weight while maintaining accuracy. Focal-EIoU Loss improves convergence and accuracy, and the CBAM attention mechanism enhances feature extraction and detection performance	YOLOv9	Zhang (2024), 2024
"A performance comparison of YOLOv8 models for traffic sign detection in the Robotaxi-full scale autonomous vehicle competition"	To develop a traffic sign recognition system using YOLOv8 for real-time detection and classification of objects, and to compare the performance of YOLOv8 models for traffic sign detection within the Robotaxi-full framework	YOLOv8 was trained on a dataset of traffic sign images to develop a model capable of accurately recognizing and classifying various types of traffic signs	YOLOv8	Soylu and Soylu (2024), 2024
"A Small Object Detection Algorithm for Traffic Signs Based on Improved YOLOv7"	A traffic sign detection algorithm for small-sized signs was proposed, built upon the enhanced YOLOv7 model, improving its ability to accurately identify and classify small traffic signs in challenging environments. Their effectiveness was validated on a public dataset	YOLOv7 was enhanced with a small target detection layer incorporating ACmix modules for improved feature extraction via self-attention and convolution. ODConv replaced standard convolution in the neck layer, boosting feature extraction. Additionally, the NWD metric was introduced to reduce sensitivity to positional deviations, improving small traffic sign detection accuracy	YOLOv7	Li et al. (2023), 2023
"Local Regression Based Real-Time Traffic Sign Detection using YOLOv6"	The capabilities of YOLOv6 are combined with an optimized Logistic Regression (LR)-based classifier, delivering improved accuracy and performance tailored to resource-constrained environments	YOLOv6 is well-suited for applications on embedded systems and smartphones, where power efficiency is a critical constraint	YOLOv6	Kaur and Singh (2022), 2022

Table 4 (continued)

Title of paper	Description of work	Purpose and YOLO usage	Version	Refs. and Year
"Two Novel Models for Traffic Sign Detection Based on YOLOv5s"	Based on the YOLOv5s architecture, two novel traffic sign detection models, YOLOv5-DH and YOLOv5-TDHSA, have been proposed. Their effectiveness was validated on a public dataset	Proposed YOLOv5-DH and YOLOv5-TDHSA, enhancing YOLOv5s with transformer self-attention modules, decoupled heads, small-object detection layers, and adaptive anchors. YOLOv5-DH focuses on accuracy and convergence, while YOLOv5-TDHSA improves small object detection	YOLOv5	Bai et al. (2023), 2023
"Small-object detection based on YOLOv5 in autonomous driving systems"	Investigated and refined YOLOv5 for improved detection of small objects such as traffic signs and traffic lights, tested on BDD100K, TT100K, and DTLD datasets	Introduced architectural changes to the popular YOLOv5 model to improve its performance in the detection of small objects without sacrificing the detection accuracy of large objects	YOLOv5	Mahaur and Mishra (2023), 2023
"Deep convolutional neural network for enhancing traffic sign recognition developed on YOLO V4"	Analyzed YOLO V4 and YOLO V4-tiny with SPP for better feature extraction in traffic sign recognition	Compared improving traffic sign recognition performance by integrating SPP into YOLO V4 backbones	YOLOv4	Dewi et al. (2022), 2022
"Real-Time Detection Method for Small Traffic Signs Based on YOLOv3"	YOLOv3 was enhanced for real-time localization and classification of small traffic signs, effectively utilizing fine-grained features in the lower layers for accurate object detection. Experiments were conducted using the TT100K dataset	MSA_YOLOv3, was proposed for real-time localization and classification of small traffic signs. It incorporates image mixup for data augmentation, integrates a multi-scale spatial pyramid pooling block into Darknet53 for comprehensive feature learning, and enhances YOLOv3's feature pyramid with a bottom-up augmented path for accurate object localization	YOLOv3	Zhang et al. (2020), 2020
"A real-time Chinese traffic sign detection algorithm based on modified YOLOv2"	A Chinese traffic sign detection algorithm based on YOLOv2 was presented. To effectively detect small traffic signs, input images are divided into dense grids to capture finer feature maps. Experiments were conducted using the Chinese Traffic Sign Dataset (CTSD)	YOLOv2 was modified by adding multiple 1×1 convolutional layers in the intermediate layers and reducing the convolutional layers in the top layers to reduce computational complexity. To enhance small traffic sign detection, input images are divided into dense grids, capturing finer feature maps	YOLOv2	Zhang et al. (2017), 2017

Table 4 (continued)

Title of paper	Description of work	Purpose and YOLO usage	Version	Refs. and Year
"YOLO-Based Traffic Sign Recognition Algorithm"	A modified YOLO architecture for traffic sign recognition, which categorizes and preprocesses traffic signs before inputting them into an optimized CNN for finer classification. Tested on a German traffic sign dataset, the algorithm improves speed and accuracy, outperforming baseline models for traffic sign recognition systems	A classical YOLO architecture was modified by enhancing the pooling layer, which reduces the spatial resolution of the input and eliminates redundant information, improving efficiency and focus on relevant features	YOLO	Li et al. (2022), 2022

Luo et al. (2021) leveraged YOLOv5 in conjunction with ResNet50 to enhance chest abnormality detection, demonstrating the algorithm's proficiency in identifying subtle medical conditions (Luo et al. 2021). Similarly, Wu et al. (2022) developed Me-YOLO, an adapted version of YOLOv5, to improve the detection of medical personal protective equipment, highlighting the model's adaptability to varied medical use cases (Wu et al. 2022). Moreover, advancements like the CSFF-YOLOv5 by Zhao et al. (2024) introduced modifications for better feature fusion, significantly boosting the detection accuracy in femoral neck fracture cases (Zhao et al. 2024). This specificity is further explored by Goel and Patel (2024), who enhanced YOLOv6 for lung cancer detection using an advanced PSO optimizer, underscoring the potential of YOLO algorithms in facilitating early disease diagnosis and treatment (Goel and Patel 2024). Additionally, the extension of YOLOv6 by Norkobil Saydirasulovich et al. (2023) for improved fire detection in smart city environments exemplifies the algorithm's versatility beyond traditional medical applications, proving its efficacy in diverse environmental conditions (Norkobil Saydirasulovich et al. 2023). Each of these developments not only enhances specific medical diagnostic processes but also paves the way for integrating these advanced object detection systems into broader healthcare applications, as illustrated by the innovative uses of YOLOv7 and YOLOv8 in detecting whole body bone fractures and enhancing hospital efficiency (Zou and Arshad 2024; Salinas-Medina and Neme 2023). These studies collectively demonstrate the significant advancements brought by YOLO in the healthcare sector, ensuring more precise, efficient, and versatile diagnostic solutions.

Recent versions such as YOLOv7, YOLOv8 and YOLOv9 have been effectively demonstrated across a variety of healthcare applications. Razaghi et al. (2024) utilized YOLOv8 for the innovative diagnosis of dental diseases, highlighting its precision in identifying dental pathologies (Razaghi et al. 2024). Similarly, Pham and Le (2024) leveraged YOLOv8 for the detection and classification of ovarian tumors from ultrasound images, showcasing the model's adaptability to different medical imaging modalities (Pham and Le 2024). Krishnamurthy et al. (2023) applied custom YOLO architectures to enhance object detection capabilities during endoscopic surgeries, illustrating the potential of YOLO in surgical settings (Krishnamurthy et al. 2023). Furthermore, Palanivel et al. (2023) discussed the application of YOLOv8 in cancer diagnosis through medical imaging, further cementing YOLO's role in critical healthcare applications (Palanivel et al. 2023).

Continuing with advancements, Karaköse et al. (2024) introduced CSFF-YOLOv5, an improved YOLO model for femoral neck fracture detection, utilizing advanced feature fusion techniques (Karaköse et al. 2024). Inui et al. (2023) demonstrated YOLOv8's effectiveness in detecting elbow osteochondritis dissecans in ultrasound images, which supports its use in orthopedic diagnostics (Inui et al. 2023). Bhojane et al. (2023) employed YOLOv8 for detecting liver lesions from MRI and CT images, underscoring the algorithm's capability across various imaging technologies (Bhojane et al. 2023). Additionally, Zhang et al. (2023) developed an improved detection model for microaneurysms using YOLOv8, which illustrates continuous enhancements in YOLO's application to highly specific medical tasks (Zhang et al. 2023).

Table 5 illustrates the different uses of YOLO versions in security and surveillance:

5.3 Security and surveillance

In the ever-evolving field of security systems, YOLO's application extends to detecting unauthorized entries and identifying potential threats swiftly, thereby bolstering security measures (Majeed et al. 2022; Aboah et al. 2021). Recent YOLO models such as YOLOv6 enhanced security and surveillance applications by improving detection accuracy through deeper network layers that process images with greater precision (AFFES et al. 2023). Meanwhile, YOLOv7 offers advanced customization options that allow security systems to be finely tuned to specific surveillance needs, enhancing the adaptability and effectiveness of threat detection (AFFES et al. 2023; Cao and Ma 2023). These YOLO versions support high-resolution video feeds, ensuring that security personnel can engage with real-time data to make informed decisions quickly. Further advancements in surveillance systems are embodied by YOLOv8 and YOLOv9, which introduce significant innovations in deep learning for security applications (Chatterjee et al. 2024; Sandhya and Kashyap 2024; Tran et al. 2024). YOLOv8's architecture is designed to handle complex environments where traditional surveillance systems may fail, such as varying lighting and weather conditions. This version's robust performance in diverse scenarios enhances its utility in comprehensive security strategies. On the other hand, YOLOv9 pushes the boundaries of speed and accuracy, providing unparalleled real-time analysis and detection capabilities. Its deployment in surveillance systems ensures that even the subtlest anomalies are detected, reducing the likelihood of security breaches. The integration of recent versions of YOLO such as YOLOv8 and YOLOv9 into security frameworks not only streamlines operations but also ensures a proactive approach to threat management, keeping public and private spaces safer across the globe (Bakirci and Bayraktar 2024a, b; Shoman et al. 2024c).

The application of YOLO models in surveillance and security systems highlights their pivotal role in enhancing real-time response and precision. Majeed et al. (2022) investigated the effectiveness of a YOLOv5-based security system within a real-time environment, underscoring its capability to significantly improve operational efficiency in dynamic settings. Similarly, AFFES et al. (2023) conducted a comparative study across YOLOv5, YOLOv6, YOLOv7, and YOLOv8, focusing on their performance in intelligent video surveillance systems. Their analysis demonstrated the incremental improvements in detection accuracy and processing speed, crucial for real-time security applications. Further advancing the field, Cao and Ma (2023) utilized a refined YOLOv7 model to enhance campus security through improved target detection capabilities, highlighting the model's precision in identifying potential threats in densely populated environments. Chatterjee et al. (2024) introduced a YOLOv8-based intrusion detection system specifically tailored for physical security and surveillance, which significantly contributes to safeguarding assets and individuals by detecting unauthorized entries or activities effectively. Additionally, Sandhya and Kashyap (2024) employed YOLOv8 for real-time object-removal tampering localization in surveillance videos, a crucial technology for maintaining the integrity of video evidence and ensuring the reliability of surveillance feeds. Together, these studies showcase the robustness of YOLO architectures in addressing diverse and complex security challenges, providing substantial improvements in both the efficacy and efficiency of surveillance operations.

Recent studies have significantly leveraged advanced YOLO models to enhance surveillance and security across various domains. Bakirci and Bayraktar (2024a) discussed optimizing ground surveillance for aircraft monitoring using YOLOv9, highlighting its efficacy

Table 5 Studies on YOLO applications in healthcare and medicine, emphasizing object detection for diagnostic imaging and real-time medical analysis

Title of paper	Description of work	Purpose and YOLO usage	Version	Refs. and Year
"Development of an AI-Supported Clinical Tool for Assessing Mandibular Third Molar Tooth Extraction Difficulty Using Panoramic Radiographs and YOLO11 Sub-Models"	Develops an AI tool using YOLO11 sub-models to evaluate mandibular third molar extraction difficulty from panoramic radiographs, incorporating different YOLO11 sizes for scenario-based evaluations	Enhances dental practice by providing an AI tool for accurate and reliable assessment of tooth extraction difficulty, facilitating improved decision-making and patient management	YOLO11	Akdoğan et al. (2025), 2025
"A Machine Learning-Based Model for the Detection of Skin Cancer Using YOLOv10"	Presents a machine learning model for skin cancer detection using YOLOv10, involving preprocessing, augmentation, and training on two datasets for different skin conditions	Aims to improve early detection and survival rates for skin cancer through advanced YOLO-based image detection	YOLOv10	Ali et al. (2024), 2024
"Efficient Skin Lesion Detection using YOLOv9 Network"	Utilized YOLOv9 for advanced skin lesion detection, leveraging deep learning to enhance diagnostic accuracy and speed	Developed improved skin lesion identification using YOLOv9, showcasing significant advances in detection performance	YOLOv9	Ju and Cai (2023), 2023
"Fracture detection in pediatric wrist trauma X-ray images using YOLOv8 algorithm"	Employed YOLOv8 with data augmentation on the GRAZPEDWRI-DX dataset for detecting fractures in pediatric wrist X-ray images	Enhanced fracture detection in pediatric wrist trauma using YOLOv8, achieving superior mAP compared to previous versions. Designed an app for surgical use	YOLOv8	Ju and Cai (2023), 2023
"Chapter 4 - Medical image analysis of masses in mammography using deep learning model for early diagnosis of cancer tissues"	Utilizes YOLOv7 to detect and diagnose cancerous tissues in mammogram images, leveraging advancements in deep learning for early cancer detection	Aims to enhance early detection of breast cancer using YOLOv7, improving diagnostic accuracy with deep learning integration. Performance measured by Precision, Recall, and F1-score	YOLOv7	Julia et al. (2024), 2024
"Improving YOLOv6 using advanced PSO optimizer for weight selection in lung cancer detection and classification"	Enhanced YOLOv6 with Particle Swarm Optimization for weight optimization in lung cancer detection from CT scans	Utilized advanced PSO to optimize YOLOv6 for higher accuracy in detecting lung cancer, significantly outperforming previous methods on the LUNA 16 Dataset	YOLOv6	Goel and Patel (2024), 2024
"One-Stage methods of computer cision object detection to classify carious lesions from smartphone imaging"	Utilized YOLO v5, YOLO v5X, and YOLO v5M to detect and classify carious lesions from smartphone images	Aimed to automate caries detection with enhanced accuracy using YOLO. mAP, P, and R metrics validated performance	YOLOv5, YOLOv5X, YOLOv5M	Salahin et al. (2023), 2023
"An Improved Method of Polyp Detection Using Custom YOLOv4-Tiny"	Customized YOLOv4-tiny with Inception-ResNet-A block for enhanced detection of polyps in wireless endoscopic images	Developed to improve the detection performance of polyp detection using a modified YOLOv4-tiny. Demonstrated significant performance improvement	YOLOv4-Tiny	Doniyorjon et al. (2022), 2022

Table 5 (continued)

Title of paper	Description of work	Purpose and YOLO usage	Version	Refs. and Year
"Detection of dental caries in oral photographs taken by mobile phones based on the YOLOv3 algorithm"	Utilized YOLOv3 for detecting dental caries from mobile phone images, employing image augmentation and enhancement for improved accuracy	Enhanced detection and diagnosis of dental caries using YOLOv3, with evaluation of diagnostic precision, recall, and F1-score across different datasets	YOLOv3	Ding et al. (2021), 2021
"Automatic thyroid nodule recognition and diagnosis in ultrasound imaging with the YOLOv2 neural network"	Employed YOLOv2 for automatic detection and diagnosis of thyroid nodules in ultrasound images, enhancing diagnostic precision	Compared AI performance with radiologists using YOLOv2, showing improved accuracy and specificity in thyroid nodule diagnosis. ROC curve analysis confirms effectiveness	YOLOv2	Wang et al. (2019), 2019
"Real-Time Facial Features Detection from Low Resolution Thermal Images with Deep Classification Models"	Developed a method to localize facial features from low-resolution thermal images by modifying existing deep classification networks for real-time detection	Demonstrates how spatial information can be restored and utilized from classification models for facial feature detection, significantly reducing dataset preparation time while maintaining high precision	Custom Deep Classification Model and YOLO	Kwaśniewska et al. (2018), 2018

Table 6 Studies on YOLO usage in security and surveillance, for real-time threat detection and enhanced monitoring to improved safety measures

Title of paper	Description of work	Purpose and YOLO usage	Version	Refs. and Year
"Assessment of YOLO11 for Ship Detection in SAR Imagery Under Open Ocean and Coastal Challenges"	Evaluated YOLO11's performance for ship detection in challenging SAR imagery, distinguishing between open ocean and coastal scenarios	Aims to enhance maritime surveillance and safety by leveraging the improved detection capabilities of YOLO11 in both open ocean and coastal environments	YOLO11	Bakirci and Bayraktar (2024), 2024
"Deploying YOLOv10 for Affordable Real-Time Handgun Detection"	Investigated the deployment of YOLOv10 on cost-effective hardware like Raspberry Pi for real-time handgun detection in various scenarios, analyzing detection parameters and model performance	Enhances public safety by enabling affordable, real-time firearm detection technology in everyday environments, using YOLOv10 to adapt AI capabilities for low-cost devices	YOLOv10	Žigulić et al. (2024), 2024
"YOLOv9-Enabled Vehicle Detection for Urban Security and Forensics Applications"	Implements YOLOv9 for aerial vehicle detection via UAVs, enhancing urban security and forensic capabilities	Focus on utilizing YOLOv9 for real-time vehicle monitoring, facilitating efficient law enforcement and forensic analysis in urban settings	YOLOv9	Bakirci and Bayraktar (2024b), 2024
"SC-YOLOv8: A Security Check Model for the Inspection of Prohibited Items in X-ray Images"	Developed a custom YOLOv8 model for X-ray image analysis to detect prohibited items. Enhanced model accuracy using a novel backbone structure and data augmentation	Aimed to improve security screening effectiveness and reduce error rates in detecting prohibited items. Showcases an innovative use of YOLOv8 in security applications	YOLOv8	Han et al. (2023), 2023
"Detection of Prohibited Items Based upon X-ray Images and Improved YOLOv7"	Improved YOLOv7 with spatial attention for contraband detection in X-ray images. Implemented large kernel attention mechanisms to improve texture and feature extraction to boost accuracy	Aims to automate security inspections and enhance public safety by improving prohibited item detection with modified YOLOv7. Demonstrates YOLOv7's adaptability in security systems	YOLOv7	Yuan et al. (2022), 2022
"Suspicious Activity Trigger System using YOLOv6 Convolutional Neural Network"	Implements YOLOv6 to detect and classify suspicious activities in CCTV footage, enhancing home surveillance systems. Utilizes deep learning to automatically trigger alerts, improving response times and security effectiveness	Aims to reduce property theft by integrating YOLOv6 into home security systems to auto-detect suspicious behavior and alert users. Demonstrates YOLOv6's effectiveness in real-world security applications	YOLOv6	Awang et al. (2023), 2023
"Real-time Object Detection for Substation Security Early-warning with Deep Neural Network based on YOLO-V5"	Utilizes YOLO-v5 to enhance substation security by detecting multiple threats like fire, unauthorized entry, and vehicle misplacement in real-time. Combines deep learning with video surveillance to reduce the need for extra hardware	Designed to improve substation security management without costly additional equipment by detecting various security threats simultaneously using YOLO-v5. Demonstrates the application of YOLO-v5 in critical infrastructure protection	YOLOv5	Xiao et al. (2022), 2022

Table 6 (continued)

Title of paper	Description of work	Purpose and YOLO usage	Version	Refs. and Year
"Fighting against terrorism: A real-time CCTV autonomous weapons detection based on improved YOLO v4"	Improved YOLOv4 with SCSP-ResNet backbone and F-PaNet module for detecting weapons in CCTV footage, integrating synthetic and real-world data to enhance detection	Aims to bolster security and counter-terrorism efforts by accurately identifying weapons in CCTV using an advanced YOLOv4 architecture, demonstrating significant performance improvements	YOLOv4	Wang et al. (2023), 2023
"Automatic tracking of objects using improvised YOLOv3 algorithm and alarm human activities in case of anomalies"	Utilizes an enhanced YOLOv3 model to automatically track objects and alert for anomalies in live video feeds, comparing performance with CNNs and decision trees	Designed to enhance surveillance systems by detecting and alerting on anomalies like bag stealing and lock-breaking, demonstrating rapid processing and high detection accuracy	YOLOv3	Kashika and Venkatapur (2022), 2022
"Multi-Object Detection using Enhanced YOLOv2 and LuNet Algorithms in Surveillance Videos"	Employs a novel YOLOv2-LuNet combination for efficient multi-object tracking in video surveillance, enhancing feature extraction and object detection accuracy	Designed to improve real-time surveillance by enabling robust multi-object tracking in challenging conditions. Highlights the effectiveness of combined YOLOv2 and LuNet approach	YOLOv2	Mohandoss and Rangaraj (2024), 2024
"From Silence to Propagation: Understanding the Relationship between 'Stop Snitchin' and 'YOLO'"	Examines the cultural shift from 'Stop Snitchin" to 'YOLO' in urban hip-hop culture, highlighting the role of social media in promoting individualism and exceptionalism	Aims to explore how social media influences criminal behavior and public perception, applying cultural criminology to assess changes in social interactions and deviance	N/A	Smiley (2015), 2015

in real-time security applications. Similarly, Chakraborty et al. (2024) explored a multi-model approach for violence detection, incorporating YOLOv8 to improve public safety through automated surveillance. These advancements indicate a shift towards reliable and efficient security systems for complex scenarios.

Chen et al. (2024) delve into the application of an enhanced YOLOv8 model for large-scale security and low-altitude drone-based law enforcement, demonstrating its potential in managing security risks effectively. Further, Pashayev et al. (2023) utilize YOLO8 for intelligent face recognition in smart cameras, contributing to the development of smarter, more responsive surveillance technologies. Additionally, Kaç et al. (2024) investigate image-based security techniques for critical water infrastructure surveillance, employing YOLO models to ensure robust monitoring. Lastly, Gao et al. (2024) introduce an improved YOLOv8s network model for contraband detection in X-ray images, underscoring the versatility and precision of YOLO models in enhancing contraband security measures.

Recent advancements in surveillance technologies have leveraged the YOLO's capabilities, particularly in managing crowd dynamics and detecting critical events. Antony et al. (2024) explored the use of YOLOv8 alongside ByteTrack for crowd management, emphasizing the system's efficiency in improving surveillance and public safety. This integration marks a significant step towards enhancing real-time monitoring capabilities during large public gatherings. Concurrently, Zhang (2024) utilized a YOLO model to detect fire and smoke in IoT surveillance systems, showcasing the model's ability to respond swiftly to emergency situations, thus bolstering safety protocols within environments.

In security, Khin and Htaik (2024) conducted a comparative study of YOLOv8 with other models like RetinaNet and EfficientDet for gun detection, emphasizing YOLOv8's superior accuracy in detecting firearms within a custom dataset. It underlines the critical role of precise object detection to prevent potential threats. Additionally, Nkuzo et al. (2023) provided a comprehensive analysis of the YOLOv7 in detecting car safety belts in real-time, illustrating its importance in enforcing road safety measures. Moreover, Chang et al. (2023) developed an improved YOLOv7, equipped with feature fusion and attention mechanisms, tailored for detecting safety gear violations in high-risk environments like construction, to enhance workplace safety standards. Table 6 presents the various YOLO usage in security and surveillance.

5.4 Manufacturing

In the landscape of industrial manufacturing, the deployment of YOLO algorithms can significantly enhance various processes and quality assessment tasks such as the development of automated optical inspection (AOI) systems. Each iteration of the YOLO family, from YOLOv2 to YOLOv5, and beyond into the latest versions like YOLOv6 and YOLOv7, brings forward substantial improvements in detecting defects across various manufacturing domains (Hussain 2023; Ahmad and Rahimi 2022; Pendse et al. 2023; Yi et al. 2024). The high accuracy and real-time processing capabilities of YOLOv6 and YOLOv7, for instance, allow for immediate identification of production flaws, crucial for maintaining workflow efficiency on fast-paced production lines (Wang et al. 2023; Ludwika and Rifai 2024). Advancing into the domain of smart manufacturing, YOLO algorithms are pivotal in revolutionizing quality control mechanisms (Beak et al. 2023; Zhao et al. 2024). The continuous evolution from YOLOv5 to YOLOV6, YOLOv7, YOLOv8, and upto 10th version

of YOLO exemplifies the adaptation of deep learning to meet the stringent quality demands of modern manufacturing processes. These algorithms reduce the need for labor-intensive manual inspections, thereby minimizing the margin for human error and enhancing the overall speed of quality assessments (Hussain 2023; Ahmad and Rahimi 2022; Pendse et al. 2023; Yi et al. 2024; Beak et al. 2023; Zhao et al. 2024).

For instance, (Liu and Ye 2023) pioneered YOLO-IMF, an enhanced version of YOLOv8 tailored for precise surface defect detection in industrial settings, exemplifying the algorithm's efficacy in real-time environments. This refinement aims to cater to the high demands for accuracy in manufacturing sectors where defects can significantly impact quality and safety. Continuing this trend, (Wen and Wang 2024) introduced YOLO-SD, which utilizes simulated feature fusion for few-shot learning, enhancing YOLOv8's capability in detecting industrial defects under varied conditions. Similarly, (Karna et al. 2023) extended YOLOv8's utility in monitoring 3D printing processes by optimizing hyperparameters to detect faults more accurately, reflecting a targeted approach to maintaining production integrity. Li et al. (2024) adapted YOLOv8 to inspect cylindrical parts, a critical aspect of quality control in specialized manufacturing. Lastly, Hu et al. (2024) leveraged a conditioned version of YOLOv8, named Cond-YOLOv8-seg, to assess the uniformity of industrially produced materials, showcasing the model's versatility across different manufacturing scenarios. These innovations underscore the pivotal role of YOLO algorithms in driving forward the capabilities of industrial inspection systems, highlighting their impact on enhancing operational efficiency and product quality.

Additionally (Yang et al. 2024) introduced DCS-YOLOv8, a variant optimized for detecting steel surface defects, demonstrating its effectiveness in addressing the complexities of steel manufacturing. This adaptation ensures that even minor imperfections are identified, crucial for maintaining the structural integrity of steel products.Likewise, Wang et al. (2023) further refined YOLOv8 to develop BL-YOLOv8, focusing on road defect detection. This model enhances the safety and maintenance of transportation infrastructure by enabling more accurate and real-time detection of road surface anomalies. Similarly, Luo et al. (2023) presented a "Hardware-Friendly" YOLOv8 model designed for foreign object identification on belt conveyors, crucial for preventing equipment damage in materials handling. This version of YOLOv8 is tailored to perform well on the limited computational resources typical of industrial hardware systems. Finally, Wang et al. (2024) employed an improved YOLOv8 algorithm for the detection of defects in automotive adhesives, a critical quality control measure for ensuring vehicle safety and durability. These applications of YOLOv8 exemplify its adaptability and precision in industrial settings, where high accuracy and efficiency are paramount for operational success and safety compliance.

The recent advancements in YOLOv7 have paved the way for significant improvements in industrial inspection and monitoring systems. Wu et al. (2023) developed an enhanced YOLOv7 model specifically tailored for detecting objects in complex industrial equipment scenarios, highlighting its application in real-world settings (Wu et al. 2023). Similarly, Kim et al. (2022) implemented YOLOv7 in a real-time inspection system that leverages Moire patterns to detect defects in highly reflective injection molding products, demonstrating the algorithm's capability in manufacturing quality control (Kim et al. 2022). Further, Chen et al. (2023) explored the defect detection capabilities of YOLOv7 for automotive running lights, contributing to safer automotive systems through precise quality assurance techniques (Chen et al. 2023).

Hussain et al. (2022) applied domain feature mapping with YOLOv7 to automate inspections of pallet racking in storage facilities, enhancing safety and efficiency in logistics operations (Hussain et al. 2022). Zhu et al. (2023) extended YOLOv7's utility to the identification and classification of surface defects in belt grinding processes, aiding in maintaining the integrity of manufacturing workflows (Zhu et al. 2023). Lastly, Zhang et al. (2024) innovated with YOLO-RDP, a lightweight version of YOLOv7, optimized for detecting steel defects in real-time, showcasing the adaptability of YOLOv7 to resource-constrained environments and promoting sustainable manufacturing practices (Zhang 2024). Table 7 illustrates the different use of YOLO versions in the field of industrial manufacturing:

5.5 Agriculture

In agricultural environments, advanced object detection techniques such as YOLOv5 (Wang et al. 2022; Badgujar et al. 2023; Shoman et al. 2022), YOLOv6 (Bhat et al. 2023; Bist et al. 2023), YOLOv7 (Jiang et al. 2022; Kumar and Kumar 2023), YOLOv8 (Chen et al. 2023; Zhang et al. 2023) and YOLOv8 to YOLO11(Sharma et al. 2024) have proven to be instrumental in transforming traditional farming into smart, precision agriculture(Badgujar et al. 2024). YOLOv5, for example, has been adept at weed detection (Chen et al. 2023; Junior and Ulson 2021), enabling farmers to apply herbicides more effectively and economically by precisely identifying and localizing weed species amidst crops. This level of precision not only conserves resources but also mitigates the adverse environmental impact of excessive chemical usage. Furthermore, YOLOv6, YOLOv7 and YOLOv8 have enhanced capabilities in broader agricultural applications such as monitoring and analyzing crop health and growth patterns, significantly improving yield predictions and crop management strategies (Yu et al. 2024; Khalid et al. 2023; Gallo et al. 2023).

The recent introduction of YOLOv7 and YOLOv8 has further pushed the boundaries of agricultural innovation. YOLOv7 (Jia et al. 2023; Vaidya et al. 2023; Umar et al. 2024) and YOLOv8 (Zhang et al. 2023; Yue et al. 2023; Zayani et al. 2024) have been specifically refined to detect small pests and subtle disease symptoms on crops, which are often overlooked by human inspectors. Its enhanced deep learning framework allows for integrating complex image recognition tasks that facilitate early detection, thereby preventing widespread crop damage. On the other hand, YOLOv8 has made significant strides in fruit and other crop canopy detection tasks. Its application in orchards for detecting fruits such as apples and apple tree branches (Ma et al. 2024) supports optimal harvesting by determining the right stage of fruit maturity. This technique helps maximize the harvest quality and ensures that the fruits are picked at their nutritional peak, thereby enhancing their market value. The application of these advanced YOLO models including YOLOv5, YOLOv6, YOLOv7, and YOLOv8 represents a leap toward a more sustainable and efficient agricultural sector.

Recent studies have demonstrated the efficacy of YOLO-based models in enhancing various aspects of smart farming and agricultural automation solutions. Junos et al. (2021) optimized a YOLO-based object detection model to improve crop harvesting systems, showcasing the potential to boost yield and reduce labor costs (Junos et al. 2021). Zhao et al. (2024) extended this application to real-time object detection combined with robotic manipulation, further aligning agricultural practices with advanced automation technologies (Zhao et al. 2024). Chen et al. (2021) developed an apple detection method using a

Table 7 Studies on YOLO applications in the manufacturing industry, focusing on real-time defect detection and process optimization for improved efficiency

Title of paper	Description of work	Purpose and YOLO usage	Version	Refs. and Year
"Automated Dual-Side Leather Defect Detection and Classification Using YOLOv11: A Case Study in the Finished Leather Industry"	Explores leather defect detection on both the grain and flesh sides using YOLOv11, with a custom-designed light chamber to enhance quality control and reduce waste	Aims to improve defect detection accuracy and leather utilization in the leather industry by utilizing advanced AI capabilities of YOLOv11 for dual-side analysis	YOLOv11	Banduka et al. (2024), 2024
"A Novel YOLOv10-Based Algorithm for Accurate Steel Surface Defect Detection"	Presents YOLOv10n-SFDC, a new system improving steel surface defect detection by integrating DualConv, SlimFusionCSP modules, and Shape-IoU loss function for enhanced accuracy	Enhances steel manufacturing processes by offering a more accurate and efficient defect detection system, demonstrating significant improvements over traditional methods	YOLOv10	Liao et al. (2025), 2025
"An Improved YOLOv9 and Its Applications for Detecting Flexible Circuit Boards Connectors"	Enhances YOLOv9 with Multi-scale Dilated Attention and Deformable Large Kernel Attention for detecting defects in FPC connectors, improving feature capture and boundary adaptation	Aims to address challenges in automatic defect detection of flexible circuit board connectors by significantly enhancing detection precision and computational efficiency with modified YOLOv9	YOLOv9	Huang et al. (2024), 2024
"YOLO-IMF: An Improved YOLOv8 Algorithm for Surface Defect Detection in Industrial Manufacturing Field"	Proposes an enhanced YOLOv8, YOLO-IMF, for surface defect detection on aluminum plates. Replaces CIOU with EIOU loss function to better handle small and irregularly shaped targets, achieving significant improvements in precision	Demonstrates YOLOv8's extended applicability in industrial settings by enhancing accuracy and defect detection capabilities	YOLOv8	Liu and Ye (2023), 2023
"YOLOv7-SiamFF: Industrial Defect Detection Algorithm Based on Improved YOLOv7"	Introduces YOLOv7-SiamFF, an advanced defect detection framework employing YOLOv7 with Siamese network enhancements for superior defect identification and background noise suppression	Enhances industrial defect detection by integrating attention mechanisms and feature fusion modules, achieving higher accuracy in pinpointing defect locations	YOLOv7	Yi et al. (2024), 2024
"A Novel Finetuned YOLOv6 Transfer Learning Model for Real-Time Object Detection"	Enhances real-time object detection by integrating a transfer learning approach with a pruned and finetuned YOLOv6 model, significantly boosting detection accuracy and speed	Focuses on improving YOLOv6 for efficient object detection in embedded systems, using advanced pruning techniques for reduced model size without sacrificing performance	YOLOv6	Gupta et al. (2023), 2023
"Real-time Tool Detection in Smart Manufacturing Using YOLOv5"	Utilizes YOLOv5 for advanced real-time tool detection in manufacturing environments, optimizing object detection capabilities for precise tool localization	Aims to enhance smart manufacturing by leveraging YOLOv5 for accurate and real-time detection of various tools, contributing significantly to Industry 4.0 initiatives	YOLOv5	Zendehdel et al. (2023), 2023

Table 7 (continued)

Title of paper	Description of work	Purpose and YOLO usage	Version	Refs. and Year
"Efficient Automobile Assembly State Monitoring System Based on Channel-Pruned YOLOv4"	Implements a channel-pruned YOLOv4 algorithm to optimize monitoring in automobile assembly, enhancing detection speed without compromising accuracy	Designed to streamline assembly monitoring in industrial environments, showcasing YOLOv4's utility in enhancing operational efficiency and deployment readiness	YOLOv4	Jiang et al. (2024), 2024
"YOLO V3 + VGG16-based Automatic Operations Monitoring in Manufacturing Workshop"	Utilizes a combined YOLO V3 and VGG16 framework to recognize and monitor industrial operations accurately for Industry 4.0 manufacturing workshops	Aims to enhance production efficiency and quality by automating action analysis and process monitoring using advanced YOLO V3 and VGG16 technologies	YOLO V3, VGG16	Yan and Wang (2022), 2022
"Improvements of Detection Accuracy by YOLOv2 with Data Set Augmentation"	Employs YOLOv2 with an innovative data set augmentation method to enhance the detection accuracy and confidence in identifying defective areas in industrial products	Seeks to optimize defect detection and visualization on production lines, demonstrating YOLOv2's effectiveness with limited data augmentation options	YOLOv2	Arima et al. (2023), 2023

tailored YOLOv4 algorithm, specifically designed to support harvesting robots operating in complex environments, which significantly enhances the precision and efficiency of fruit picking (Chen et al. 2021).

Further contributions include work by Nergiz (2023), who utilized YOLOv7 to enhance strawberry harvesting efficiency, providing practical solutions for small to medium-sized enterprises in the agricultural (NERGİZ 2023). Wang et al. (2024) focused on planning harvesting operations in large strawberry fields using a deep learning-based image processing method, demonstrating the scalability of YOLO for larger agricultural operations (Wang et al. 2024). Lastly, Zhang et al. (2023) introduced DCF-YOLOv8, an improved algorithm for agricultural pests and diseases detection by aggregating low-level features, which helps in early detection and management of crop health (Zhang et al. 2023). These studies collectively illustrate the transformative impact of YOLO-based models in modernizing agricultural practices, ensuring higher productivity and sustainability.

In orchard automation, the YOLO object detection models have been specifically pivotal in enhancing the accuracy and efficiency of fruit detection (Chen et al. 2022; Mirhaji et al. 2021; Sapkota et al. 2024c), flower identification (Wu et al. 2020; Wang et al. 2021; Khanal et al. 2023), and automated harvesting processes (Xiao et al. 2023; Junos et al. 2021; Yijing et al. 2021). These models adeptly identify and classify fruits at various stages of ripeness, detect flowers and other canopy objects such as branches with high precision, and facilitate efficient harvesting operations. The development of YOLO models has introduced significant improvements that cater specifically to the challenges of agricultural environments. For instance, YOLOv5's introduction of multi-scale predictions improved the detection of small and clustered objects like flowers and young fruits, which are critical during the early stages of crop yield management (Zhang et al. 2024). As the models advanced, YOLOv7 and YOLOv8 incorporated better segmentation techniques, which enhanced the differentiation between fruit types and maturity stages, critical for targeted harvesting (Zhou et al. 2023; Xiuyan and ZHANG 2023).

In addition to their application in fruit detection and harvesting, YOLO object detection models are increasingly vital across other agricultural practices such as pruning (Sapkota et al. 2024a), thinning (Sapkota et al. 2024c), pollination and harvest (He et al. 2021) management. In pruning, YOLO models facilitate the accurate identification of non-fruiting branches, aiding in the automation of pruning tasks to optimize plant health and productivity. For thinning processes, these models can distinguish between fruit clusters, allowing for precise thinning that improves fruit size and quality at harvest. Furthermore, YOLO models are being adapted to recognize pollination patterns, helping to monitor and enhance pollination efficiency, which is crucial for maximizing crop yield. Each advancement in YOLO technology contributes to more sustainable and effective agricultural operations, underscoring the critical role of AI-driven technologies in modern farming practices.

Moreover, recent iteration, YOLOv9 have leveraged advanced algorithms with spatial pyramid pooling and attention mechanisms, which have refined the detection capabilities in plant disease detection (Boudaa et al. 2024). Boudaa et al. (2024) performed a comparative study on different important versions of YOLO (v5, v8 and v9) on a real-world dataset for tomato plant disease detection and suggested that YOLOv9 outperforms YOLOv5 and YOLOv8. A study on weed species control was presented by Sharma et al. (2024), where a comparative analysis of different YOLO versions (v8, v9, v10, and 11) and Faster R-CNN was conducted. The study utilized an annotated image database containing five weed spe-

cies: cocklebur (Xanthium strumarium L.), dandelion (Taraxacum officinale), common waterhemp (Amaranthus tuberculatus), Palmer amaranth (Amaranthus palmeri), and common lambsquarters (Chenopodium album L.).

It is also noted that autonomous driving has been an active area of research and development in agriculture as well (Andreyanov et al. 2022; Jung et al. 2020), similar to self-driving cars. YOLO models have played crucial roles in advancing this technology for farming for example in peach crop (Xu and Rai 2024). Object detection based on YOLO has enhanced the navigation systems of autonomous agricultural vehicles, enabling them to operate efficiently in diverse field conditions. By accurately detecting and classifying field boundaries, obstacles such as rocks and trees, and other critical elements like rows of crops and water sources, YOLO-powered systems significantly improve the precision and safety of operations. This capability is particularly vital in precision agriculture, where exactness in planting, fertilizing, and treatment applications can substantially impact crop health and yield. The integration of YOLO models into agricultural drones and robots also supports tasks such as crop monitoring, disease detection, and targeted pesticide application, further automating and optimizing farm management practices (Alibabaei et al. 2022).

Table 8 illustrates the different use of YOLO versions in the field of Agriculture:

6 Challenges and limitations

6.1 Challenges and limitations of each version

YOLOv12:

- Computational efficiency and performance trade-offs: YOLOv12 introduces the Area Attention (A^2) module and FlashAttention mechanism, which, while theoretically efficient, present practical challenges. Empirical studies demonstrate a reduction in inference speed (30 FPS compared to YOLO11's 40 FPS) and an increase in training time by approximately 20% (Tian et al. 2025). The effectiveness of these attention mechanisms varies with hardware architecture, potentially limiting performance gains on certain systems. Moreover, YOLOv12 requires lower confidence thresholds (0.3−0.4) for effective detection, which may increase false positive rates. These factors collectively contribute to the ongoing challenge of optimizing the trade-off between detection accuracy and real-time performance, particularly in resource-constrained environments and complex detection scenarios.
- Detection limitations and environmental sensitivity: Despite architectural advancements, YOLOv12 continues to face challenges in accurately detecting small objects and objects in crowded scenes, a limitation persistent from previous iterations. This is primarily due to three factors: 1) The downsampling process in YOLO architectures can lead to loss of fine-grained details, particularly affecting small object detection (Diwan et al. 2023), 2) Limited contextual information in a single forward pass makes it difficult for the model to differentiate between closely packed or overlapping objects in crowded scenes (Ji et al. 2023; Fang et al. 2019), and 3) The use of fixed anchor boxes with predefined scales and aspect ratios may not optimally capture objects with highly variable sizes or unusual shapes (Sirisha et al. 2023).

Table 8 Studies on YOLO usage in agriculture, emphasizing automated crop monitoring, pest detection, and yield estimation for enhanced productivity

Title of paper	Description of work	Purpose and YOLO usage	Version	Refs. and Year
"Improved YOLOv12 with LLM-Generated Synthetic Data for Enhanced Apple Detection and Benchmarking Against YOLOv11 and YOLOv10"	Evaluates YOLOv12 for apple detection in commercial orchards using LLM-generated synthetic datasets and benchmarks it against YOLOv11 and YOLOv10	Demonstrates that YOLOv12 achieves the highest precision (0.916), recall (0.969), and mAP@50 (0.978), surpassing YOLOv11 and YOLOv10 in accuracy while YOLO11 remains the fastest. Highlights efficiency in processing speeds and cost-effectiveness by reducing the need for extensive manual data collection	YOLOv12	Sapkota and Karkee (2025), 2025
"Synthetic meets authentic: Leveraging llm generated datasets for YOLO11 and YOLOv10-based apple detection through machine vision sensors"	Utilizes LLM-generated datasets to train YOLOv10 and YOLO11 for detecting apples, showcasing significant improvements in detection metrics and processing times	Aims to streamline data collection and enhance object detection in orchards with minimal fieldwork by employing YOLO11 and YOLOv10, demonstrating superior performance in real-world tests	YOLO11	Sapkota et al. (2024b), 2024
"Comprehensive Performance Evaluation of YOLO11, YOLOv10, YOLOv9 and YOLOv8 on Detecting and Counting Fruitlet in Complex Orchard Environments"	Conducts a thorough performance evaluation of YOLOv8, YOLOv9, YOLOv10, and YOLO11 on green fruit detection and counting across multiple configurations, highlighting in-field validation with an iPhone and machine vision sensors	Provides comparative insights on various YOLO configurations for optimizing fruitlet detection in commercial orchards, recommending YOLO11 for its speed and accuracy	YOLO11	Sapkota et al. (2024d), 2024
"YOLOv10-pose and YOLOv9-pose: Real-time Strawberry Stalk Pose Detection Models"	Introduces YOLOv10-pose and YOLOv9-pose for high-precision strawberry stalk pose detection, comparing them with previous YOLO versions to optimize agricultural automation tasks	Enhances the efficiency of robotic harvesting and other agricultural applications by providing accurate pose detection, crucial for automated operations in the agricultural industry	YOLOv10	Meng et al. (2025), 2025
"Comparing YOLOv11 and YOLOv8 for Instance Segmentation of Occluded and Non-occluded Immature Green Fruits in Complex Orchard Environment"	Evaluates YOLOv11 and YOLOv8 for instance segmentation capabilities of immature green fruits, focusing on occluded and non-occluded scenarios in orchards	Enhances understanding of YOLO11 and YOLOv8's segmentation performance, particularly their efficacy in detecting and segmenting immature green fruits amidst complex environmental conditions	YOLO11	Sapkota and Karkee (2024), 2024
"Automating Tomato Ripeness Classification and Counting with YOLOv9"	Implements YOLOv9 to automate and enhance the accuracy of classifying and counting ripe tomatoes, replacing labor-intensive visual inspections	Aims to streamline tomato ripeness monitoring and counting, to enhance agricultural productivity and quality. Utilizes YOLOv9 for high accuracy in detection	YOLOv9	Vo et al. (2024), 2024

Table 8 (continued)

Title of paper	Description of work	Purpose and YOLO usage	Version	Refs. and Year
"A Lightweight YOLOv8 Tomato Detection Algorithm Combining Feature Enhancement and Attention"	Enhances YOLOv8 for tomato detection in agriculture using depthwise separable convolution and dual-path attention gate modules. Optimizes real-time detection for robotic tomato picking	Aims to advance agricultural automation by boosting YOLOv8's efficiency and accuracy in tomato harvesting. Demonstrates improved performance over earlier YOLO versions	YOLOv8	Yang et al. (2023), 2023
"An Attention Mechanism-Improved YOLOv7 Object Detection Algorithm for Hemp Duck Count Estimation"	Implements CBAM-YOLOv7 to enhance feature extraction capabilities within YOLOv7 for precise hemp duck counting in agriculture, outperforming SE-YOLOv7 and ECA-YOLOv7 in precision and mAP	Enhances livestock management by automating duck count with advanced object detection, reducing labor and improving accuracy	YOLOv7	Jiang et al. (2022), 2022
"Detecting Crops and Weeds in Fields Using YOLOv6 and Faster R-CNN Object Detection Models"	Utilizes YOLOv6 and Faster R-CNN to detect crops and weeds for precise management	Aims to boost agricultural productivity and environmental sustainability by improving accuracy in weed detection using YOLOv6	YOLOv6	Bhat et al. (2023), 2023
"An improved YOLOv5-based vegetable disease detection method"	Enhances YOLOv5 for precise detection of vegetable diseases by upgrading CSP, FPN, and NMS modules to handle complex environmental interference	Aims to improve food security by boosting the accuracy and speed of disease detection in vegetables using an improved YOLOv5 algorithm	YOLOv5	Li et al. (2022), 2022
"Using channel pruning-based YOLO v4 deep learning algorithm for the real-time and accurate detection of apple flowers in natural environments"	Implements a channel pruned YOLOv4 model to enhance efficiency and accuracy in detecting apple flowers, supporting the development of flower thinning robots	Aims to optimize apple flower detection in orchards by applying channel pruning to YOLOv4, significantly reducing model size and improving processing speed while maintaining high accuracy	YOLOv4	Wu et al. (2020), 2020
"Fast and accurate detection of kiwifruit in orchard using improved YOLOv3-tiny model"	Enhances YOLOv3-tiny with additional convolutional kernels for improved kiwifruits detection in orchards, in occlusions and varying lighting conditions	Focus on increasing the efficiency of kiwifruit detection in dynamic orchard environments with a modified YOLOv3-tiny, demonstrating high performance	YOLOv3-tiny	Fu et al. (2021), 2021
"A Detection Method for Tomato Fruit Common Physiological Diseases Based on YOLOv2"	Implements YOLOv2 to detect and identify healthy and diseased tomato, using advanced image processing and data augmentation to enhance detection accuracy	Aims to boost tomato yield and quality control through efficient detection of physiological diseases, demonstrating the effectiveness of YOLOv2 in agriculture	YOLOv2	Zhao and Qu (2019), 2019

Table 8 (continued)

Title of paper	Description of work	Purpose and YOLO usage	Version	Refs. and Year
"A Vision-Based Counting and Recognition System for Flying Insects in Intelligent Agriculture"	Utilizes YOLO for initial detection and counting, and SVM for fine classification of flying insects, for efficient insect pest control	Demonstrates a robust, efficient system for insect monitoring, greatly enhancing accuracy and speed in pest management	YOLO, SVM	Zhong et al. (2018), 2018
"Comparative performance of YOLOv8, YOLOv9, YOLOv10, YOLO11 and Faster R-CNN models for detection of multiple weed species"	Compare the performance of YOLOv8, YOLOv9, YOLOv10, YOLO11, and Faster R-CNN algorithms in terms of speed and accuracy	Demonstrates the robust efficiency of several YOLO versions in detecting weed species using a database comprising cocklebur (Xanthium strumarium L.), dandelion (Taraxacum officinale), common waterhemp (Amaranthus tuberculatus), Palmer amaranth (Amaranthus palmeri), and common lambsquarters (Chenopodium album L.)	YOLOv8, YOLOv9, YOLOv10, YOLOv11 and Faster R-CNN	Sharma et al. (2024), 2024

These issues contribute to the model's potential struggle with objects at vastly different scales within the same image, indicating a need for improved scale-invariant feature extraction.

YOLOv11:

As the most recent addition to the YOLO series, this version must address and overcome

- Challenges in Detecting Small and Rotated Objects: Despite advancements, YOLOv11 still struggles with small, low-resolution objects and those with varied orientations. This limitation is due to its architectural constraints, which may not fully capture the complexities of such objects, leading to potential inaccuracies in detection.
- Susceptibility to Overfitting: YOLOv11 is prone to overfitting, particularly when trained on limited or homogeneous datasets. This overfitting can adversely affect the model's performance on new or varied datasets, indicating a need for improved generalization capabilities and robust training approaches.
- Computational Efficiency vs. Accuracy Trade-off: While YOLOv11 has improved computational efficiency, there remains a trade-off with accuracy, particularly in complex detection environments. This trade-off highlights the ongoing challenge of balancing speed and accuracy to support real-time applications.

YOLOv10:

- YOLOv10 has not yet seen widespread adoption in published research. Its release promises cutting-edge improvements in object detection capabilities, but the lack of extensive testing and real-world application data makes it difficult to ascertain its full potential and limitations.
- Preliminary evaluations suggest that while YOLOv10 might offer advancements in speed and accuracy, integrating it into existing systems could present challenges due to compatibility and computational demands. Potential users may hesitate to adopt this version until more comprehensive studies and benchmarks are available, which articulate its advantages over previous models.
- The expectation with YOLOv10, much like its predecessors, is that it will drive further research in object detection technologies. Its eventual widespread implementation could pave the way for addressing complex detection scenarios with higher accuracy, particularly in dynamic environments. However, as with any new technology, the adaptation phase will be crucial in understanding its practical limitations and operational challenges.

YOLOv9:

- Despite YOLOv9's enhancements in detection capabilities, it has only been featured in a handful of studies, which limits a comprehensive understanding of its performance across diverse applications. This lack of extensive validation may deter organizations from adopting it until more empirical evidence and comparative analyses establish its efficacy and efficiency over earlier versions.
- YOLOv9 significantly rectified the computational efficiency challenges faced by

YOLOv8. YOLOv9 introduces several improvements that enhance its efficiency while maintaining or improving accuracy. The model achieves a 49% reduction in parameters and a 43% reduction in computational load compared to YOLOv8, while improving accuracy by 0.6% on benchmark dataset. The introduction of the Generalized Efficient Layer Aggregation Network and Programmable Gradient Information enhances feature extraction and gradient flow, leading to improved efficiency. YOLOv9-E demonstrates a 16% reduction in parameters and a 27% reduction in FLOPs compared to YOLOv8-X, while also gaining a 1.7% improvement in mAP value.

- While YOLOv9 improves upon the speed and accuracy of its predecessors, it may still struggle with detecting small or overlapping objects in cluttered scenes. This is a recurring challenge in high-density environments like crowded urban areas or complex natural scenes in transportation and agriculture, where precise detection is critical for applications such as autonomous driving, wildlife monitoring and robotic fruit picking.
- Future developments for YOLOv9 could focus on enhancing its robustness in adverse conditions, such as varying weather, lighting, or occlusions. Integrating more adaptive and context-aware mechanisms could help in mitigating false positives and improving the reliability of the system under different operational conditions. The implementation of advanced training techniques such as federated learning could also be explored to enhance its adaptability and learning efficiency from decentralized data sources.

YOLOv8:

- YOLOv8 has shown significant improvements in object detection tasks, particularly in real-time applications. However, it continues to face challenges in terms of computational efficiency and resource consumption when deployed on lower-end hardware (Ye et al. 2024). This can limit its applicability in resource-constrained environments where deploying advanced hardware solutions is not feasible (Soylu and Soylu 2024).
- The future direction for YOLOv8 could involve optimizing its architectural design to reduce computational load without compromising detection accuracy. Enhancing its scalability to efficiently process images of varying resolutions and conditions can broaden its application scope. Moreover, incorporating adaptive scaling and context-aware training methods could potentially address the detection challenges in complex scenes, making it more robust against diverse operational challenges.

YOLOv7:

- Although YOLOv7 introduces significant improvements in detection accuracy and speed, its adoption across varied real-world applications reveals a persistent challenge in handling highly dynamic scenes. For instance, in environments with rapid motion or in scenarios involving occlusions, YOLOv7 can still experience drops in performance. The algorithm's ability to generalize across different types of blur and motion artifacts remains an area for further research and enhancement.
- The complexity of YOLOv7's architecture, while beneficial for accuracy, imposes a substantial computational burden. This makes it less ideal for deployment on edge devices or platforms with limited processing capabilities, where maintaining a balance between speed and power efficiency is crucial (Olorunshola et al. 2023; AFFES et al.

2023). Efforts to streamline the model for such applications without significant loss of performance are necessary.

- Looking forward, there is significant potential in expanding YOLOv7's capabilities through the integration of semi-supervised or unsupervised learning paradigms. This would enable the model to leverage unlabeled data effectively, a common challenge in the real-world where annotated datasets are often scarce or expensive to produce. Additionally, enhancing the model's resilience to adversarial attacks and variability in data quality could further solidify its utility in security-sensitive applications like surveillance and fraud detection.

YOLOv6:

- One of the notable challenges with YOLOv6 is its handling of scale variability within images, which can affect its efficacy in environments where objects appear at diverse distances from the camera. While YOLOv6 shows improved accuracy and speed over its predecessors, it sometimes struggles with small or partially occluded objects, which are common in crowded scenes or complex industrial environments (Norkobil Saydirasulovich et al. 2023; Li et al. 2023). This limitation can be critical in applications such as automated surveillance or advanced manufacturing monitoring.
- YOLOv6, while efficient, still requires considerable computational resources when compared to other models optimized for edge devices. Its deployment in resource-constrained environments such as mobile or embedded systems often requires a trade-off between detection performance and operational efficiency. Further optimizations and model pruning are necessary to achieve the best of both worlds-real-time performance with reduced computational demands.
- Future enhancements for YOLOv6 could focus on incorporating more advanced feature extraction techniques that improve its robustness to variations in object appearance and environmental conditions. Additionally, integrating more adaptive and context-aware learning mechanisms could help overcome some of the challenges related to background clutter and similar adversities. Enhancing the model's capacity to learn from a limited number of training samples, through techniques such as few-shot learning or transfer learning, could address the scarcity of labeled training data in specialized applications.

YOLOv5:

- YOLOv5 has made significant strides in improving detection speed and accuracy, but it faces challenges in consistently detecting small objects due to its spatial resolution constraints. This is particularly evident in fields such as medical imaging or satellite image analysis, where precision is crucial for identifying fine details. Techniques such as spatial pyramid pooling or enhanced up-sampling may be needed to increase the receptive field and improve the detection of smaller objects without compromising the model's efficiency (Benjumea et al. 2021; Jung and Choi 2022; Wang et al. 2022).
- While YOLOv5 offers faster training and inference times compared to previous versions, its deployment on edge devices is limited by high memory and processing requirements (Wu et al. 2021; Jia et al. 2023). Although optimized models like YOLOv5s provide a solution, they sometimes do so at the cost of detection accuracy. Optimizing

network architecture through neural architecture search (NAS) could potentially offer a more balanced solution, enhancing both performance and efficiency for real-time object detection applications.

- The adaptability of YOLOv5 to varied environmental conditions and different types of data distribution remains an area for development. Future research could focus on enhancing the robustness of YOLOv5 through advanced data augmentation techniques and domain adaptation strategies. This would enable the model to maintain high accuracy levels across diverse application settings, from urban surveillance to complex natural environments, effectively handling variations in lighting, weather, and seasonal changes.

YOLOv4, YOLOv3, YOLOv2 and YOLOv1:

- While YOLOv4 introduced notable enhancements in speed and accuracy, it still exhibits performance inconsistencies across different datasets, particularly with class imbalance and the detection of rare objects. The model's high computational demand also restricts its deployment on low-power devices. Continued efforts to improve model compression and increase adaptability to varying environmental conditions are essential to extend its practical utility in diverse real-world applications.
- YOLOv3 improved upon the balance of speed and accuracy, yet it struggles with small object detection due to its grid limitation. Its computational efficiency poses challenges for deployment in resource-constrained environments, prompting research towards optimization techniques to improve efficiency without sacrificing performance. Additionally, enhancing the model's robustness to environmental variations could improve its reliability for applications like autonomous driving and urban surveillance.
- Despite the incremental improvements introduced in YOLOv2, it faces challenges in detecting small objects, balancing speed with accuracy, and maintaining relevance with the advent of more capable successors. This version's reliance on a fixed grid system hampers its ability to perform in high-precision detection tasks. Future developments may shift towards adapting YOLOv2's core strengths in new architectures that enhance its spatial resolution and dynamic scaling capabilities.

For the versions of YOLO under YOLOv5, their use may decrease and discontinue in the future as newer versions are replacing the older YOLO versions in overall performance and efficiency.

- The potential for YOLOv4, YOLOv3, and YOLOv2 in future research involves exploring adaptive mechanisms that can tailor learning rates and augment data to better handle diverse operational scenarios. Integrating these models with newer technologies like model pruning and feature fusion may address existing inefficiencies and extend their applicability to a wider range of applications.
- YOLOv1 was revolutionary for its time, introducing real-time object detection by processing the entire image at once as a single regression problem. However, it faces significant challenges in dealing with small objects due to each grid cell predicting only two boxes and the probabilities for the classes. This structure often leads to poor performance on groups of small objects that are close together, such as flocks of birds or traffic

scenes with multiple vehicles at a distance. Improvements in subsequent models focus on increasing the number of predictions per grid and incorporating finer-grained feature maps to enhance small object detection.

- Another limitation of YOLOv1 is the spatial constraints of its bounding boxes. Since each cell in the grid can only predict two boxes and has limited context about its neighboring cells, the precision in localizing objects, especially those with complex or irregular shapes, is often compromised. This challenge is particularly evident in medical imaging and satellite image analysis, where the exact contours of the objects are crucial. Advances in convolutional neural network designs and cross-layer feature integration in later versions seek to address these drawbacks.
- Although YOLOv1 laid the groundwork for real-time object detection, its direct usage has significantly diminished, with advancements in the field largely driven by more recent iterations such as YOLOv4 and beyond. These newer models have not only retained the core principles of YOLOv1 but have also introduced improved mechanisms for handling diverse object sizes and aspect ratios. Current and future research is less likely to concentrate on YOLOv1 and earlier versions like YOLOv3, but rather on advancing these later iterations or developing hybrid models that might incorporate elements of YOLOv1's architecture to benefit applications where high speed and low latency are paramount, despite potential trade-offs in detection precision and detail.
- Future iterations could focus on dynamic grid systems, lighter network architectures, and advanced scaling features to tackle the challenges of small object detection and computational limitations. These improvements could enhance their deployment in emerging areas such as edge computing, where real-time processing and low power consumption are crucial.
- As YOLO continues to evolve with newer iterations like YOLOv8 and YOLOv9, the core principles of earlier versions such as YOLOv4, YOLOv3, and YOLOv2 still hold significant value for developing hybrid models and specialized applications. The research community is increasingly focused on harnessing the rapid detection capabilities of these older versions while addressing their limitations in detection accuracy through composite and hybrid modeling strategies. This trend is evidenced by innovations that integrate YOLO with other architectures to enhance overall performance. For example, a hybrid model that combines CNNs, YOLO, and Vision Transformers (ViTs) has demonstrated enhanced detection accuracy and reduced inference times by utilizing CNNs for robust feature extraction, YOLO for quick object detection, and ViTs for capturing global context (Ali et al. 2024). Similarly, the DA-ActNN-YOLOv5 model merges YOLOv5 with advanced data augmentation and ActNN's model compression techniques to optimize both accuracy and efficiency across diverse operational environments (Zhu et al. 2021; Bashir et al. 2023)

6.2 Challenges in statistical metrics for evaluation

Threat: Evaluating YOLO detection systems requires a unique approach, as each version, from the original YOLO to the latest YOLOv12, targets different aspects of detection capability, such as speed, accuracy, or computational efficiency. For a comprehensive evaluation, it is essential to employ a diverse array of metrics, including precision, recall, GFLOPs, and model size. This approach allows for a more complete comparison and understanding

of each model's strengths and weaknesses in various real-world applications. This multi-metric evaluation is crucial to assessing the practical utility and technological advancement of the YOLO series. Future YOLO versions are expected to introduce novel evaluation metrics that capture emerging capabilities in edge computing, multi-modal fusion, and adaptive architecture optimization, necessitating an even more sophisticated evaluation framework to accurately assess their performance across diverse deployment scenarios.

Mitigation: Despite this limitation, our main premise is that the selected metrics enable us to compare various YOLO systems and adequately assess their overall effectiveness. Recognizing the inherent limitations of statistical summaries is crucial when conducting a comprehensive evaluation of detection systems across different applications. Therefore, we aim to improve the clarity and reliability of our review by openly acknowledging these potential threats to construct validity. This approach provides a more nuanced understanding of the limitations associated with various aspects of YOLO techniques for object detection in diverse domains.

Spectral versus RGB images: Beyond traditional RGB imaging, spectral features encompass a broader spectrum, including infrared, ultraviolet, and even multispectral and hyperspectral imaging. These advanced spectral techniques can significantly enhance YOLO's object detection capabilities by providing additional information not visible in the RGB spectrum. For example, hyperspectral imaging can detect subtle variations in plant health for agricultural applications or distinguish between materials based on their spectral signatures in industrial settings. This expansion into wider spectral data not only improves detection accuracy but also opens up new avenues for application-specific optimizations, reinforcing YOLO's versatility and potential across various fields.

Over the past decade, the series of YOLO models have significantly impacted various sectors, demonstrating the powerful capabilities of deep learning in real-world applications. As a pioneering object detection algorithm, YOLO has facilitated rapid advancements across diverse fields by offering high-speed, real-time detection with commendable accuracy. One of the most notable applications has been in public safety and surveillance, where YOLO models have improved the efficacy of monitoring systems, enhancing the detection of suspicious activities and ensuring public safety more efficiently. In the realm of automotive technology, YOLO has been integral in developing advanced driver-assistance systems (ADAS) (Malligere Shivanna and Guo 2024), contributing to object detection that supports collision avoidance systems and pedestrian safety. Furthermore, YOLO has transformed the healthcare sector by accelerating medical image analysis, enabling quicker and more accurate detection of pathologies which is critical for diagnostics and treatment planning. In industrial settings, YOLO has optimized quality control processes by identifying defects in manufacturing lines in real-time, thereby reducing waste and increasing production efficiency. Additionally, in the retail sector, YOLO has supported inventory management through automated checkouts and stock monitoring, enhancing customer experience and operational efficiency, whereas in agriculture, YOLO has played a key role to enhance timely crop stress detection, pest localization and precision crop management while improving worker health and safety.

7 Future directions in object detection with YOLO

7.1 YOLO deployment on edge and IoT devices

The deployment of YOLO on edge devices unlocks several promising avenues for future research and development. One potential direction involves enhancing the algorithm's efficiency and accuracy for even more constrained environments, such as ultra-low-power Microcontrollers and embedded systems. This can be achieved through further optimization techniques, including model pruning, quantization, and the development of specialized hardware accelerators. Additionally, integrating YOLO with advanced communication protocols, edge computing frameworks and IoT devices could facilitate more seamless collaboration between edge devices and centralized cloud services, enhancing the overall system performance and scalability. Exploring the integration of YOLO with other AI-driven functionalities, such as anomaly detection and predictive analytics, may unlock new applications in areas like healthcare, smart cities, and industrial automation. As edge computing continues to evolve, the adaptation of YOLO to support federated learning paradigms could ensure the data privacy while enabling continuous learning and improvement of object detection models. These future directions will not only expand the capabilities of YOLO but also contribute significantly to the advancement of intelligent edge computing systems (Yang et al. 2022; Ghaziamin et al. 2024; Hussain et al. 2022; Zhang 2024).

7.2 YOLO and embodied artificial intelligence

Embodied Artificial Intelligence (EAI) refers to AI systems integrated with physical entities or bodies, enabling them to interact with the real world in a natural and human-like manner (Pfeifer and Iida 2004). Incorporating YOLO into these systems significantly enhances their sensory capabilities, allowing for more efficient and accurate interaction with the physical environment. Applications of YOLO in EAI include autonomous vehicles, drones, robots (Sanket 2021), human-robot interaction (Wang et al. 2024). Additionally, it plays a significant role in healthcare, particularly with robotic surgical assistants (Lakshmipathy et al. 2024), among other innovative uses (Li et al. 2024).

8 Expanding YOLO object detection into broader AI domains

8.1 YOLO and artificial general intelligence

Artificial General Intelligence (AGI) refers to an intelligent agent with human-level or higher intelligence, capable of solving a variety of complex problems in diverse domains (Pande et al. 2024; Qu et al. 2024). In this context, an AGI system would need to integrate object detection capabilities, similar to those provided by YOLO, with other essential cognitive functions, such as advanced natural language understanding, reasoning, and decision-making. This fusion will enable the system to handle a broad spectrum of tasks in real-time, adapting to dynamic environments and complex scenarios effectively, thus advancing the current AI systems towards achieving a true AGI.

8.2 YOLO integration with large language models

One effective way to advance AGI capabilities is to integrate YOLO with Large Language Models (LLMs) by seamlessly merging advanced visual data interpretation with sophisticated natural language understanding, reasoning, and contextual awareness. This fusion would allow the AGI to not only recognize and analyze objects in real-time but also engage in meaningful interactions with stakeholders (or end users), making more informed decisions and adapting to complex tasks with greater autonomy and precision. This synergy would enable AGI systems to operate in complex environments handling multi-modal inputs simultaneously, such as navigating through an environment using visual cues identified by YOLO while interpreting and acting on spoken commands through capabilities provided by LLMs (Rouhi et al. 2025; Sapkota et al. 2024e, b). Such integration is expected to lead to a highly versatile and intelligent system, capable of performing real-time, multi-faceted operations across diverse application domains. By combining advanced visual recognition with robust language processing, it brings us a step closer to realizing true AGI, where the system can autonomously adapt, learn, and perform complex tasks with human-like flexibility and reasoning. Future YOLO versions, such as YOLOv13, YOLOv14, YOLOv15, and beyond, are anticipated to advance toward AGI, integrating enhanced reasoning, adaptability, and autonomous learning capabilities beyond traditional object detection.

9 YOLO and environmental impact

Training and retraining YOLO is extremely energy-intensive, leading to substantial energy and water consumption, as well as significant carbon dioxide emissions. This environmental impact underscores concerns about the sustainability of AI development, emphasizing the urgent need for more efficient practices to reduce the ecological footprint of large-scale model training (Xu et al. 2024; Dhar 2020).

10 Conclusion

In this comprehensive review, we explored the evolution of the YOLO models from the most recent YOLOv12 to the inaugural YOLOv1, including alternative versions of YOLO as YOLO-NAS, YOLO-X, YOLO-R, DAMO-YOLO, and Gold-YOLO. This retrospective analysis covered a decade of advancements, highlighting theapplied use of each version and their respective impacts across five critical application areas: autonomous vehicles and traffic safety, healthcare and medical imaging, security and surveillance, manufacturing, and agriculture. Our review outlined the significant enhancements in detection speed, accuracy, and computational efficiency that each iteration brought, while also addressing the specific challenges and limitations faced by earlier versions. Furthermore, we identified gaps in the current capabilities of YOLO models and proposed potential directions for future research, such as trade-off between detection speed versus accuracy, handling small and overlapping Objects, and generalization across diverse datasets and domains. Predicting the trajectory of YOLO's development, we anticipate a shift towards multimodal data processing, leveraging advancements in large language models and natural language processing to enhance

object detection systems. This fusion is expected to broaden the utility of YOLO models, enabling more sophisticated, context-aware applications that could revolutionize the interaction between AI systems and their environments using Generative AI and multi-modal LLMs. Thus, this review not only serves as a detailed chronicle of YOLO's evolution but also sets a prospective blueprint for its integration into the next generation of technological innovations.

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Data availability No datasets were generated or analysed during the current study.

Declarations

Conflict of interest The authors declare no conflict of interest.

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